



CONTAINER REFRIGERATION UNITS





TOPICS

- **Models**
- **Components**
- **Refrigeration Systems**
- **Controllers**
- **Options**

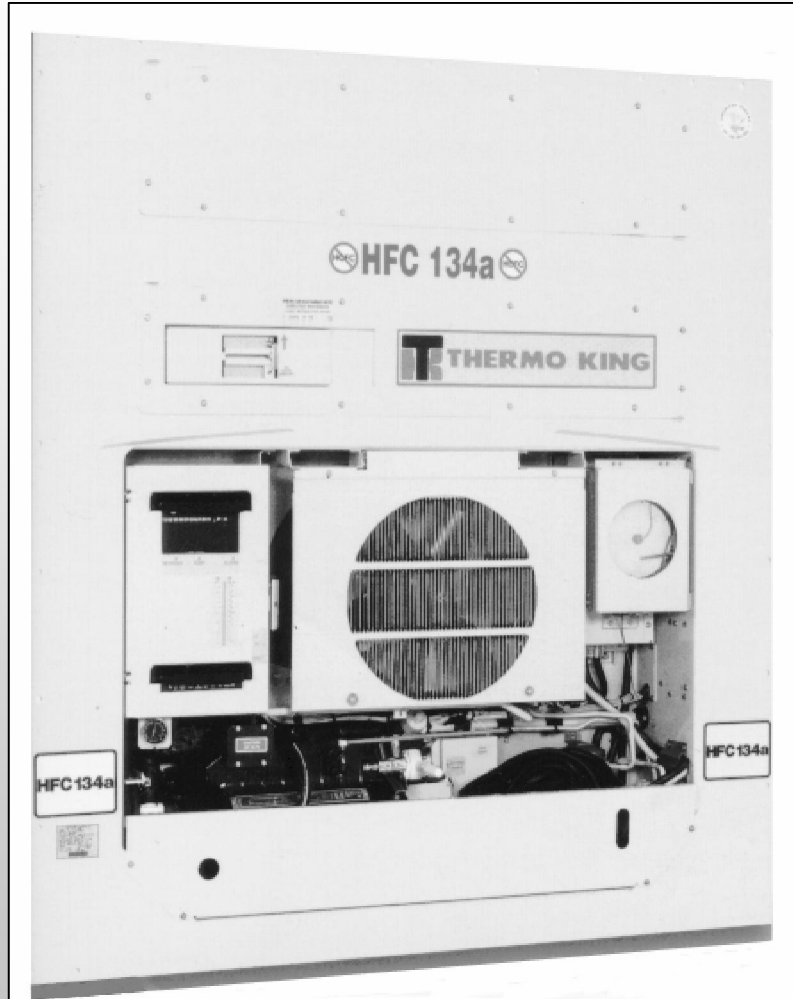


MODELS

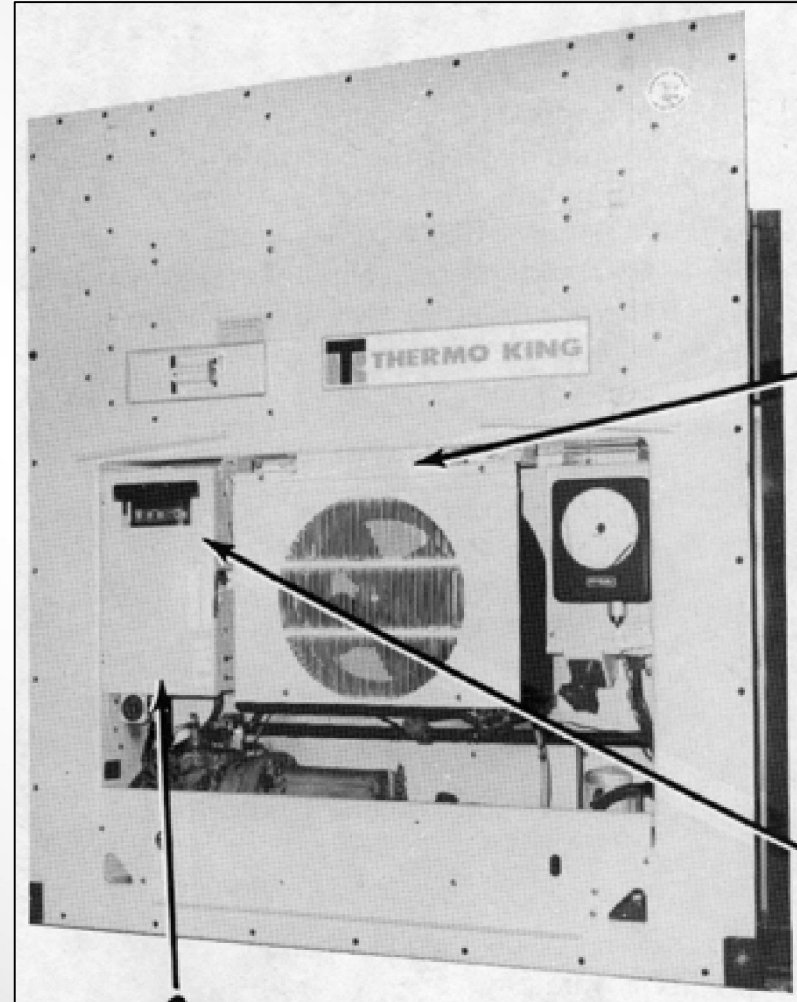
<u>MODEL</u>	<u>COMP.</u>	<u>REFRIG.</u>	<u>CONTROLLER</u>
CF-II / CF-III	X430/X426	R134a / R12	μP-A+, μP-A, μP, μC
CF-II	3DS/3DF	R134a / R12	μP-A+, μP-A
CF-II M39	2DB	R404A	μP-A+
CRR (TNE)	3DS / 3DF	R134a	μP-D / MPC2000 / ID
TNE (Sabroe)	Scroll	R22	MPC2000
CSR	Scroll	R404A	μP-D
CRR PS / PS+ 3DS		R134a	MP 3000
CSR PS / PS+ Scroll		R404A	MP 3000



CF-II

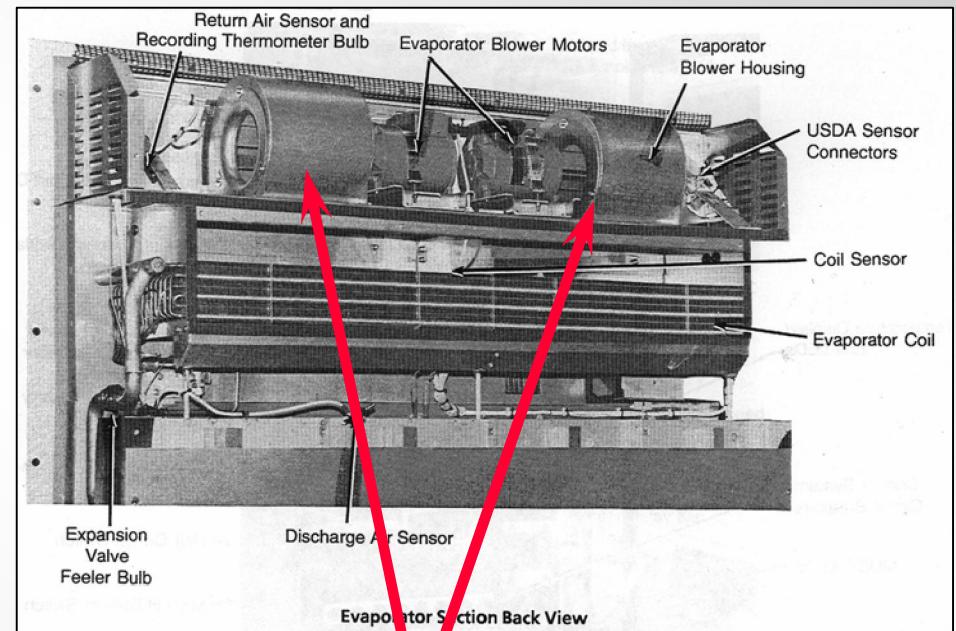
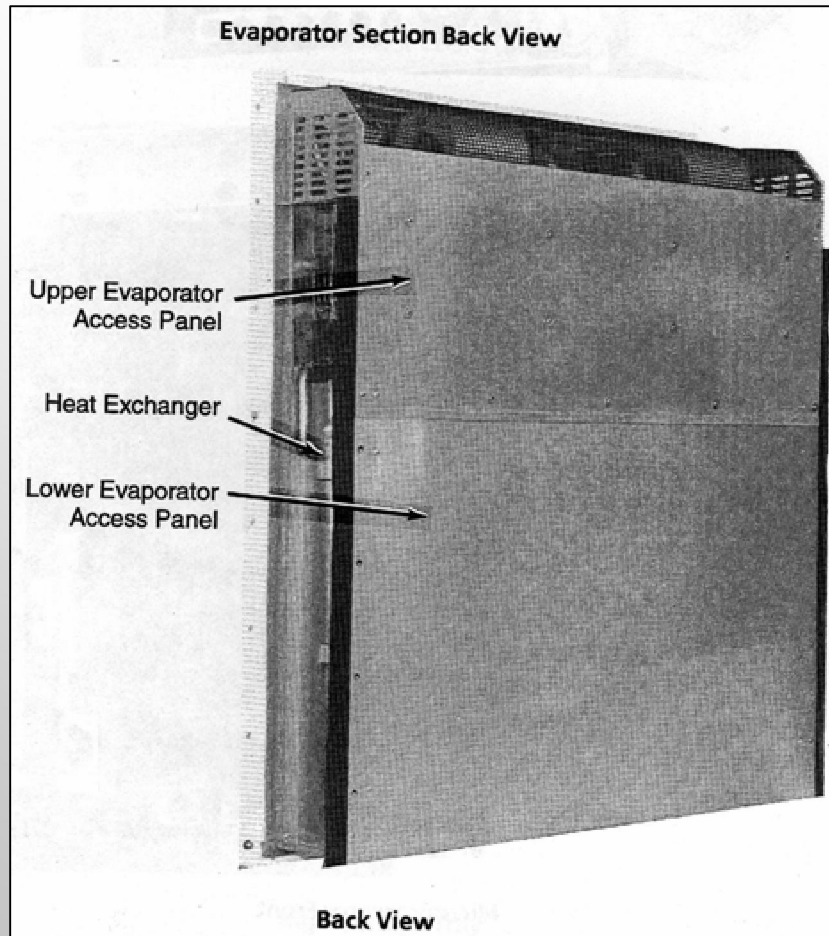


CF-III



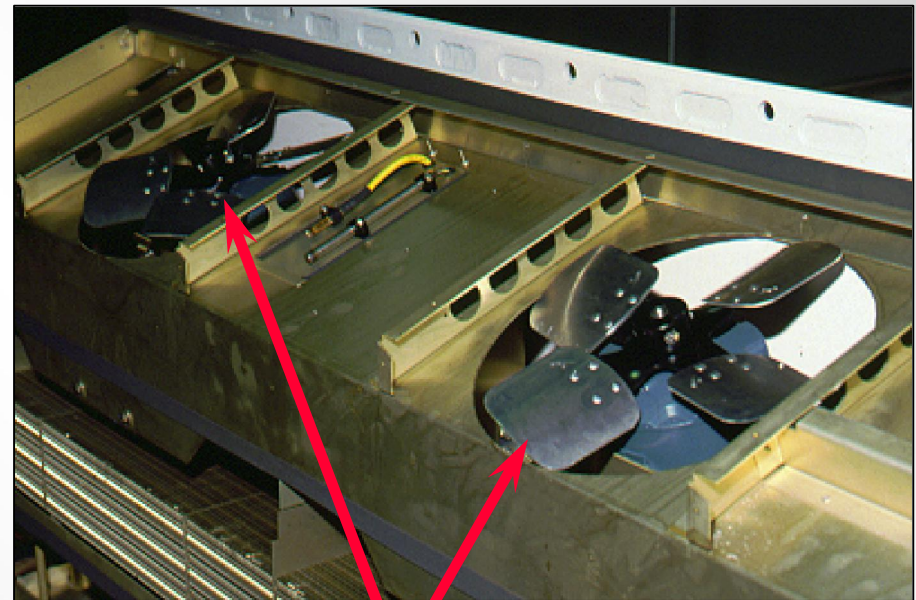
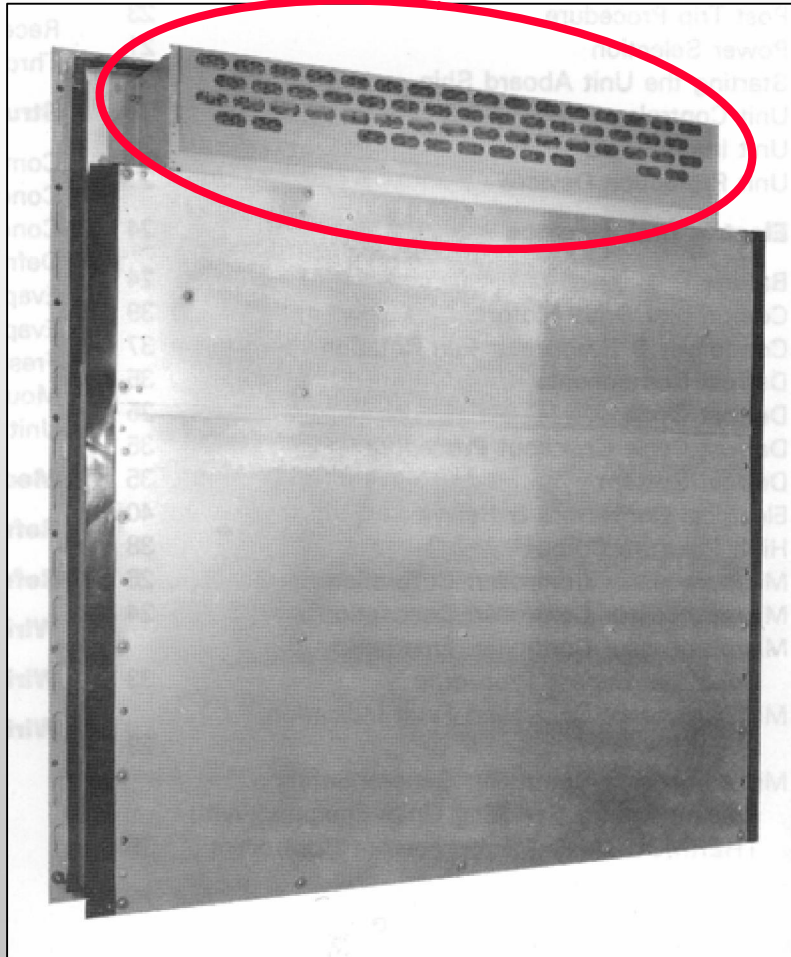


CF-II EVAP. SECTION



Blowers / Wheels

CF-III EVAP. SECTION

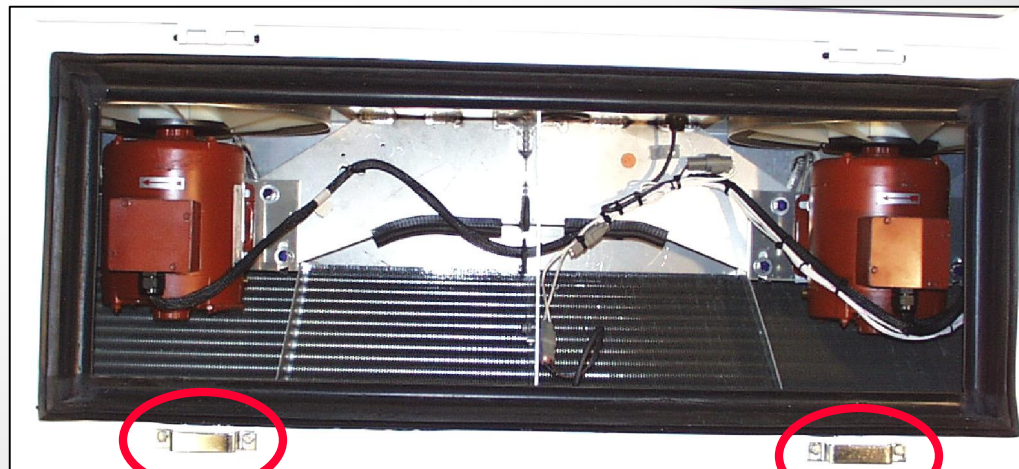


Propeller Fans

EVAP. FAN CONFIGS.

2 Fans 'Standard'

- CSR & CRR PS / PS Models

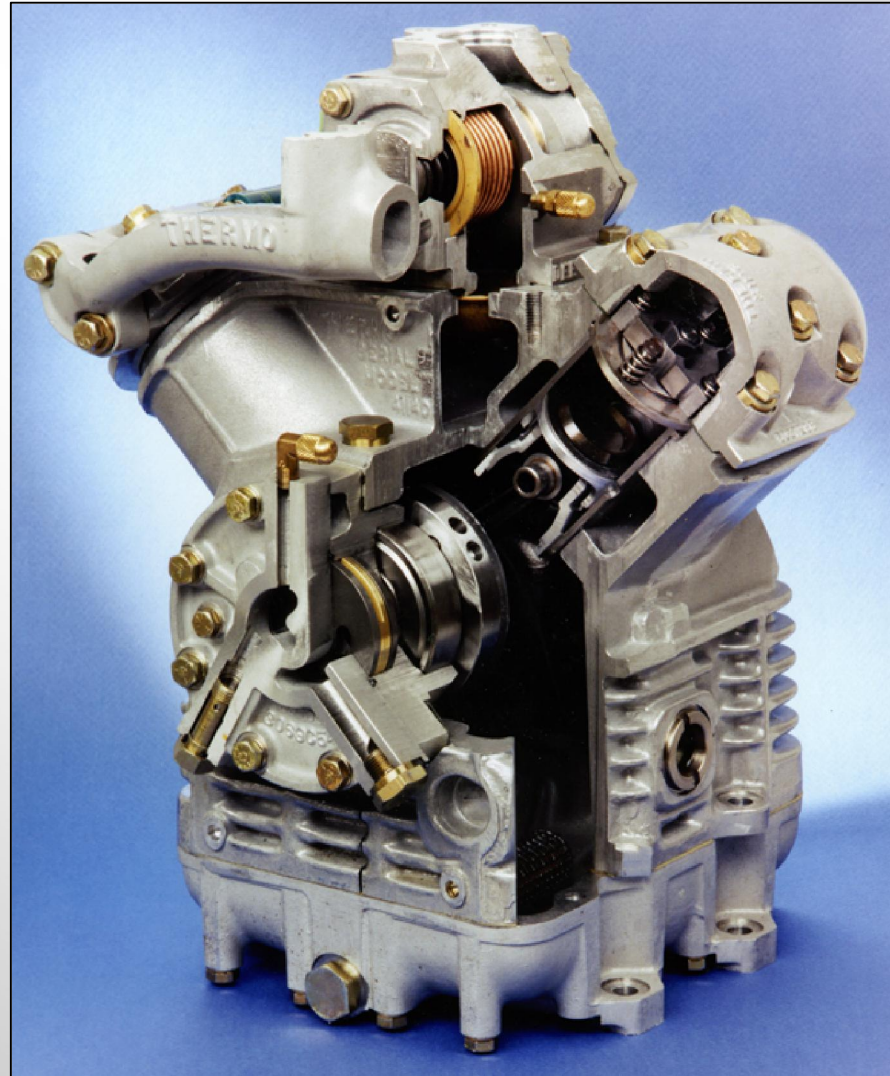


3 Fans - Slimline Models



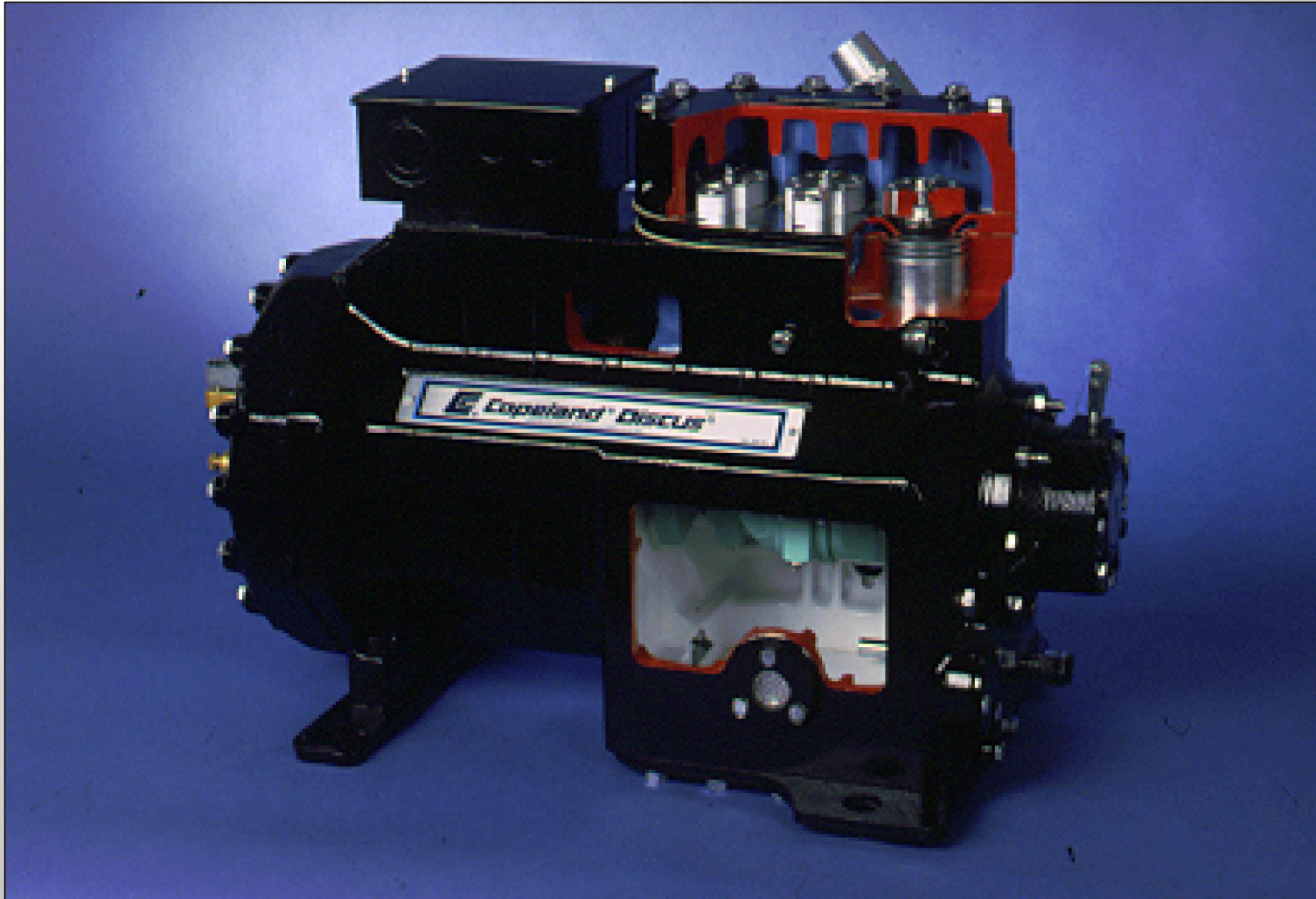


X430 / X426 COMPRESSOR



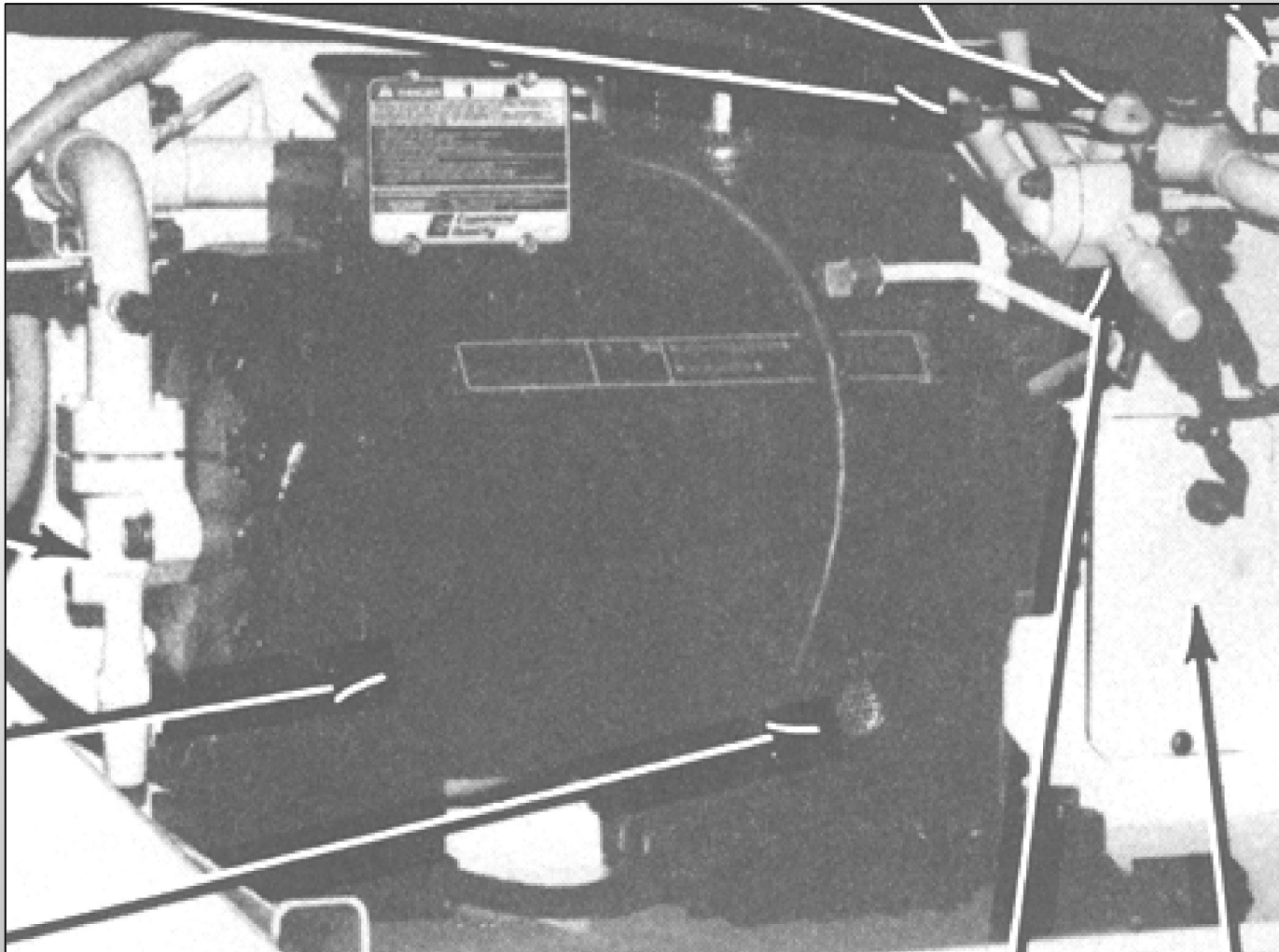


3DS / 3DF COMPRESSOR

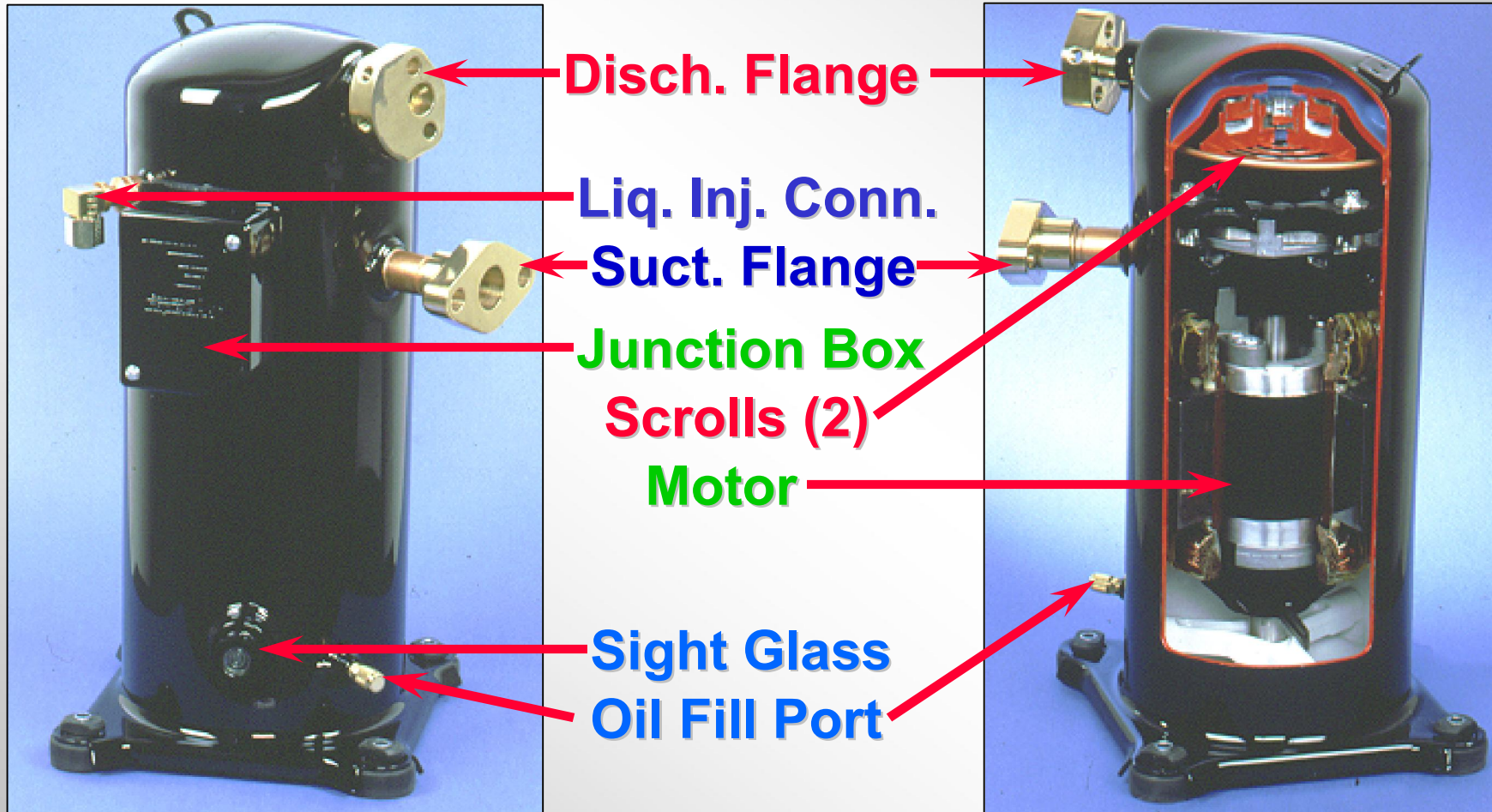




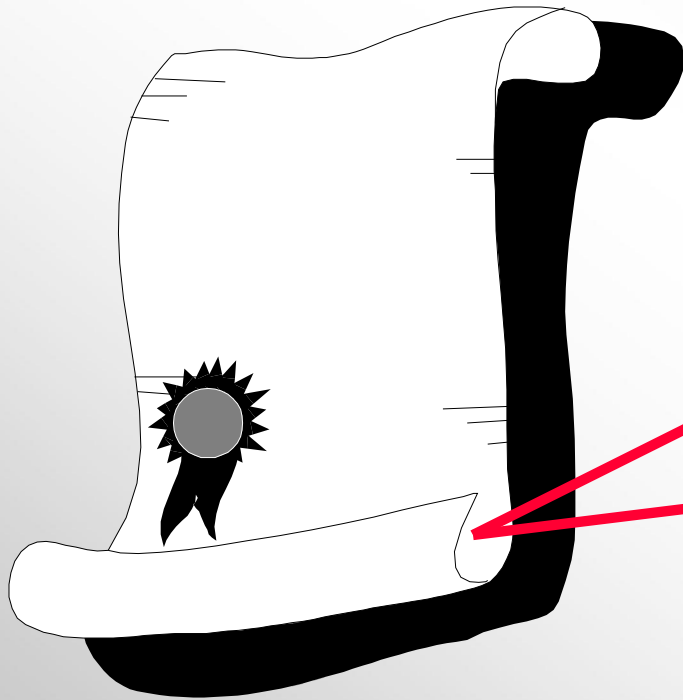
2DB COMPRESSOR



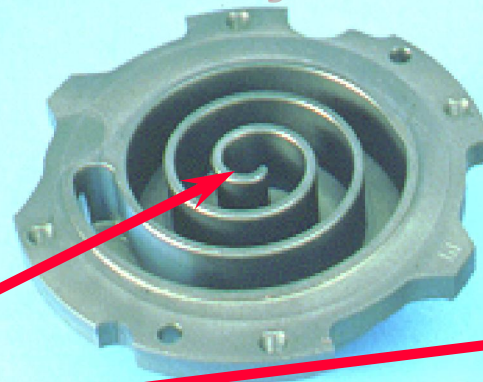
'FLANGED' SCROLL



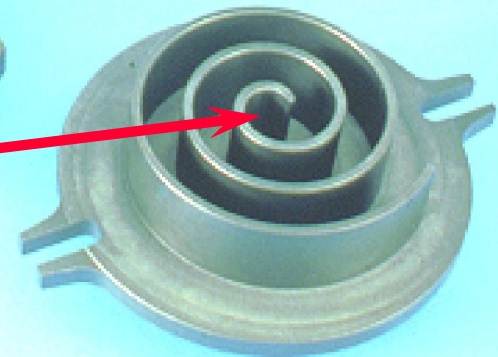
WHAT'S A SCROLL?



'Stationary' Scroll



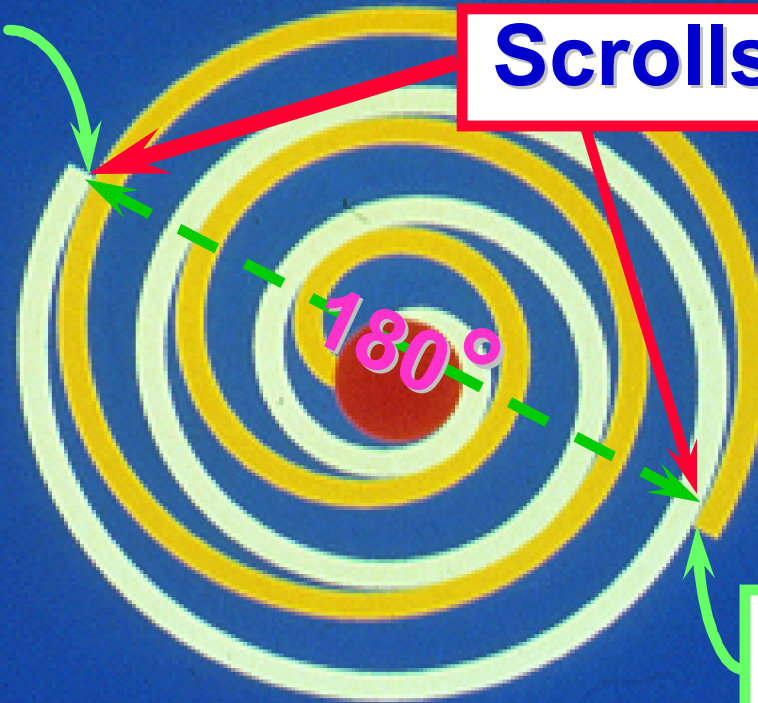
'Orbiting' Scroll



Scroll Design

Seal

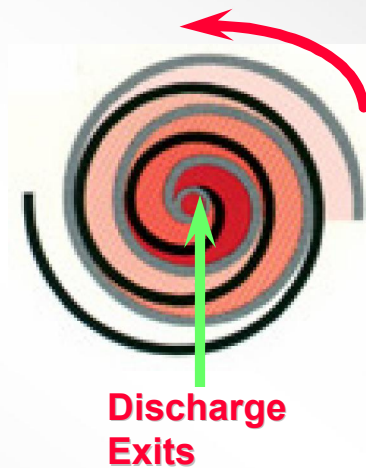
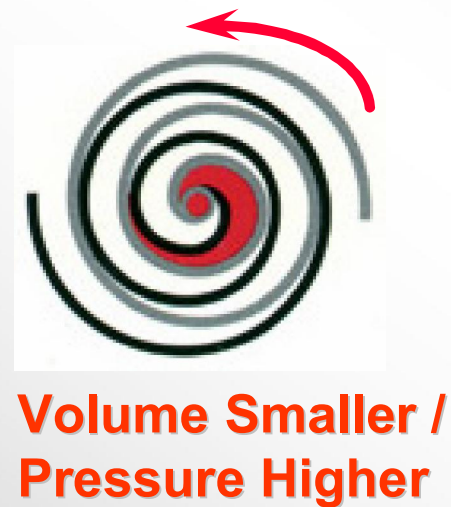
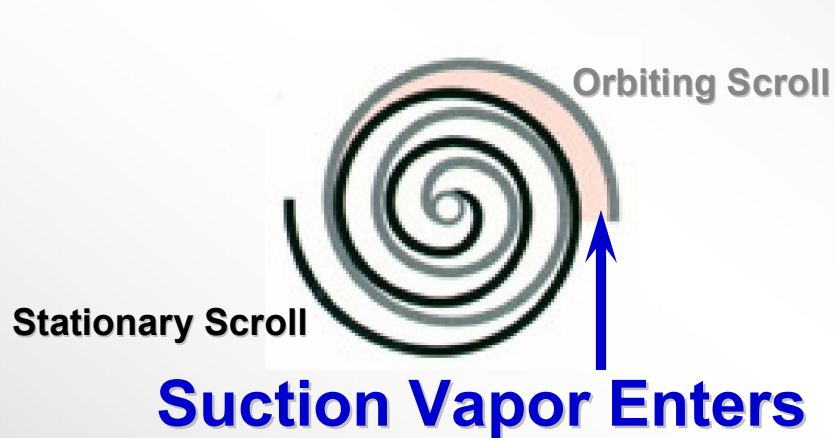
Scrolls Opposed



Seal



COMPRESSION CYCLE

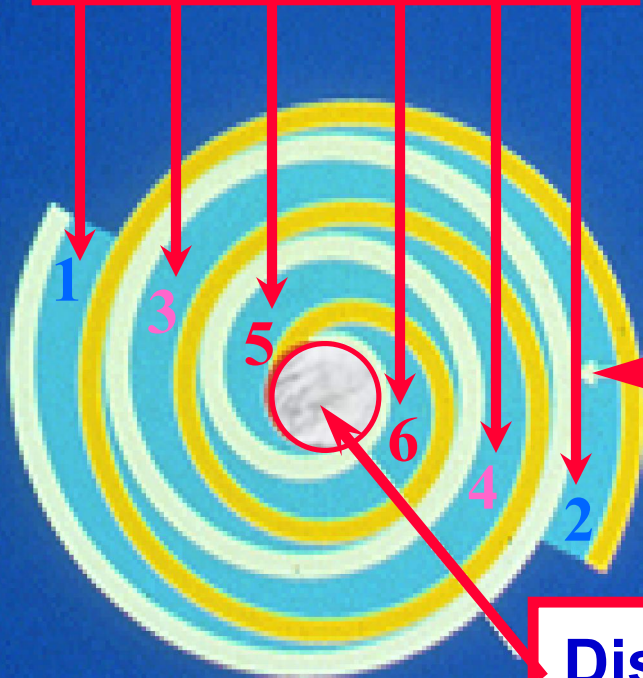




SCROLL ADVANTAGES

SCROLL

6 Chambers



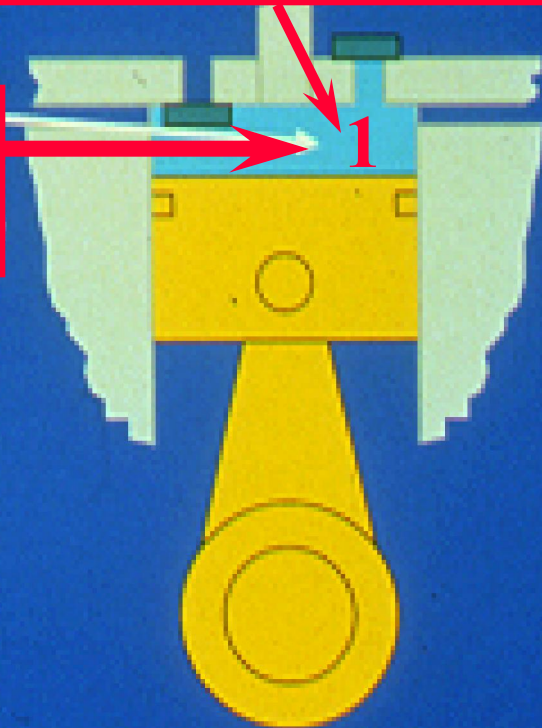
Discharge

Suction

Suction & Discharge

RECIP. PISTON

One Chamber





SCROLL 'COMPLIANCE'

- **What is Scroll Compliance?**
 - Scroll's Ability to Separate from One Another in the Presence of Liquid and / or Debris
- **Types of Compliance (2)**

–Axial 

–Radial 



OTHER PS / PS+ FEATURES

Dual Supply Air Sensors (LH / RH)

- Chill Setpoints
 - < 30 F (-1 C) uses coldest sensor
 - > 32 F (0 C) uses average of both
 - Between 30 F (-1 C) & 32 F (0 C) - Uses Sliding Scale
 - Coldest to Average

Modulation Cool - Near Setpoint

- Return Air within .5 F (0.3 C) of setpoint, heaters pulse ON/OFF via 30 sec duty cycle



OTHER FEATURES CON'T

Modulation Display in Data Menu

- i.e. Display 70% = 30% Capacity Reduction

Compressor Liquid Injection - Anytime...

- Compressor Starts = 5 Min. 'On'
- Modulation < 75% (or Power limit)
- Comp. Temp. > 280 F (138 C)

Warm Gas Bypass

- Modulation < 25% = ON/OFF 30 Sec. Cycle
 - 'ON' Time Increases as Mod. Decreases



OTHER FEATURES CON'T

Evaporator Fan Control (PS Units Only)

- High / Low Speed Based on Setpoint 14F (-10C)
- Economy Mode
 - Chilled Setpoint & In Range = Low Speed
 - Frozen Setpoint & Null = OFF

Condenser Fan Control

- Low Ambient - Maintains Min. Cond. Temp.
 - Chill Loads = Min. 86 F (30 C)
 - Frozen Loads = Min. 68 F (20 C)
- Cycles ON 10 Sec. of Every 1 Min. Cycle
- > 155 F (68 C) Cond. Temp.= ON



OTHER FEATURES CON'T

Compressor 'Short Cycling' Prevention

- Comp. Starts = On for Minimum 5 Min.
- Comp. Stops = Off for Minimum 5 Min.

Defrost Initiation

- Demand or Manual - Coil Sensor < 65 F (18 C)
- Timed - Coil Sensor < 50 F (10 C)

Demand Defrost Initiation Rules

- Large Difference Between...
 - Return and Coil or...
 - Left and Right Supply or...
 - Return and Supply



OTHER FEATURES CON'T

Timed Defrost Initiation

- Starts 1 Min. Past the Hour
- Chilled Setpoints
 - Defrost Every 8 Hours if Supply Air > 41.2 F (5.1 C)
 - First Defrost @ 2.5 hrs. if Supply Air < 41.2 F (5.1 C), .5 Hrs. Added to previous time. i.e. 3, 3.5, 4, 4.5, etc. 7 hrs Max.
- Frozen loads
 - First Defrost @ 8 Hrs. 2 Hrs. added to previous time after every timed defrost. 24 hrs Max.
- Timer Resets Whenever...
 - Setpoint Change > 8.9 F (5 C)
 - Off More Than 12 Hours



OTHER FEATURES CON'T

Defrost Termination Rules

- **Chilled Loads**
 - Coil Sensor > 86 F (30 C) or...
 - Exceeds 65 F (18 C) for 15 Min.
- **Frozen Loads**
 - Coil Sensor > 65 F (18 C) or...
 - Exceeds 46 F (8 C) for 15 Min.
- **Defrost Termination Timer**
 - 60 Hz = 90 Min.
 - 50 Hz = 120 Min.
- **Pre-Cool Evap. Coil**
 - 3 Mins. Max. or Supply Air Temp. to Setpoint



OTHER FEATURES CON'T

Probe Test

- High Speed Evap. Fans Only for 5 Min.
 - Sensors w/Large Differences Discarded from Algorithm
 - If No Sensor Failure Found, Displays “Running with High Supply Difference”

Heater Operation

- Modulation Cool Near Setpoint - 30 Sec. Duty Cycle
- Heat Mode - 60 Sec. Duty Cycle
- Defrost Mode - ON ‘All the Time’

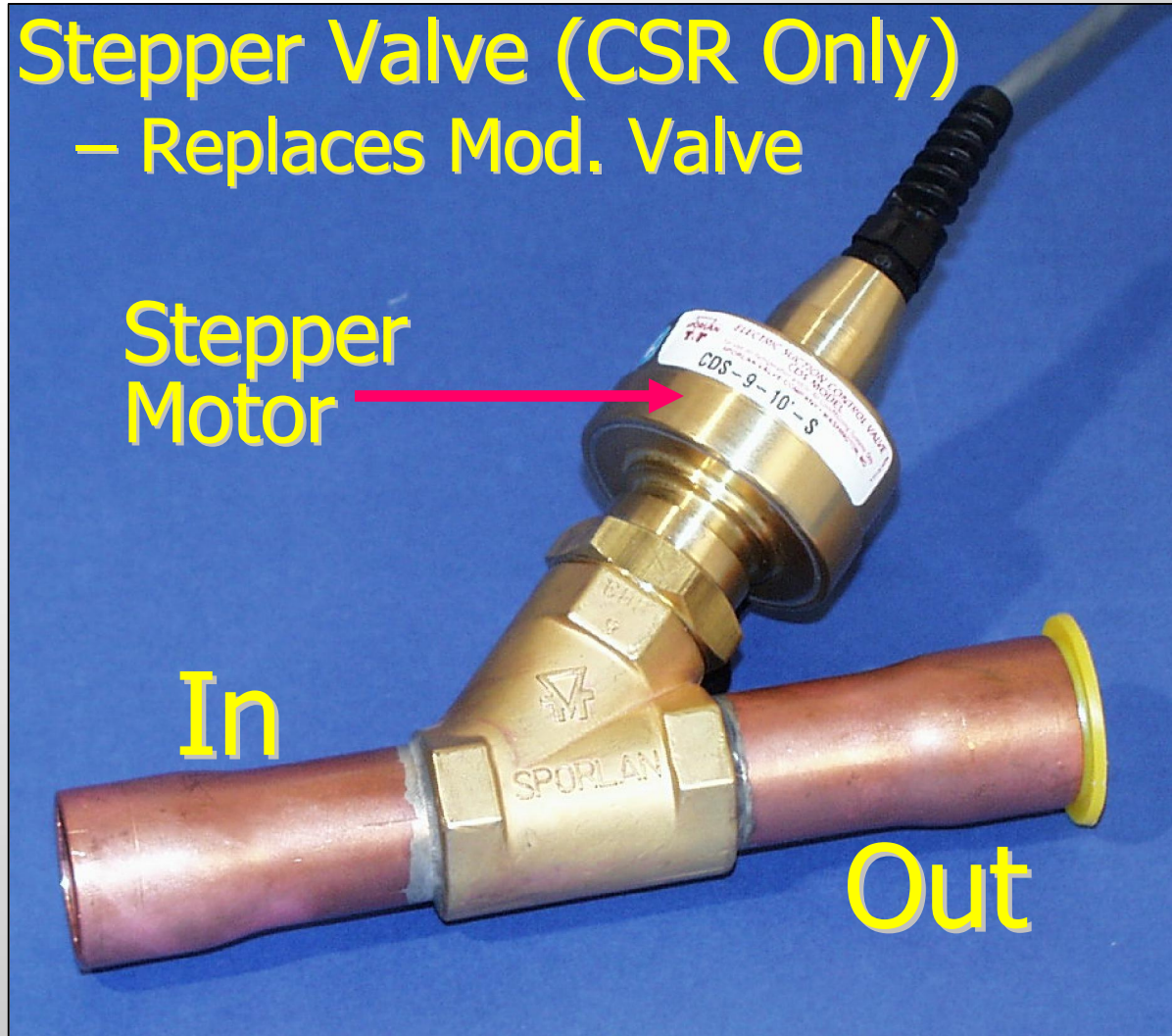
'CSR' PS / PS+

Stepper Valve (CSR Only)
– Replaces Mod. Valve

Stepper
Motor

In

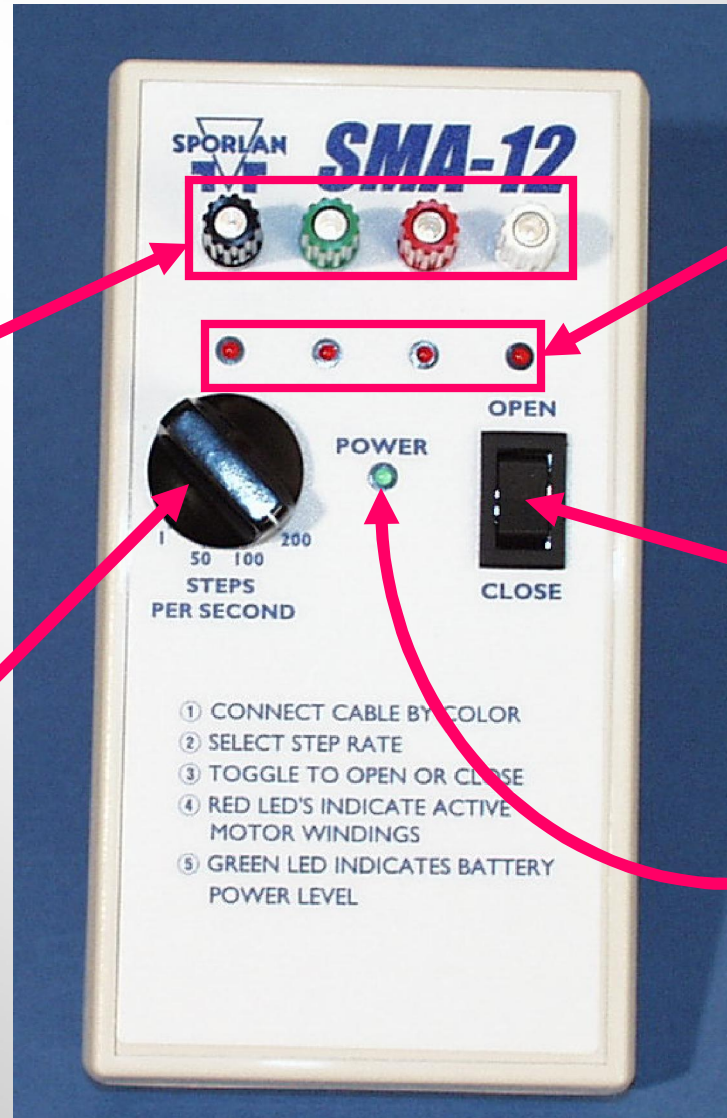
Out



TESTER



TESTER CON'T



Elect. Conn.

Motor
Winding
Indicators

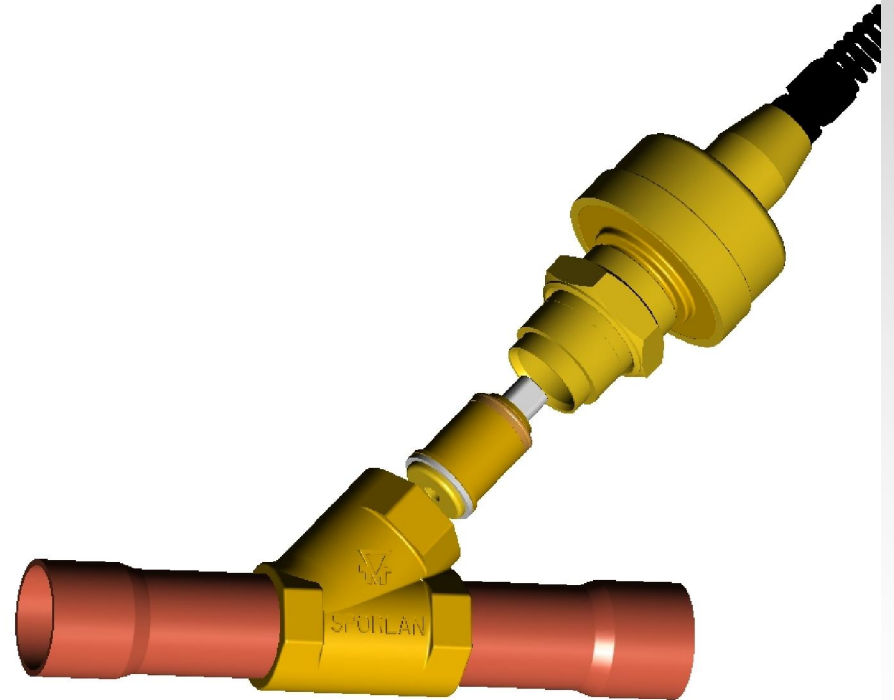
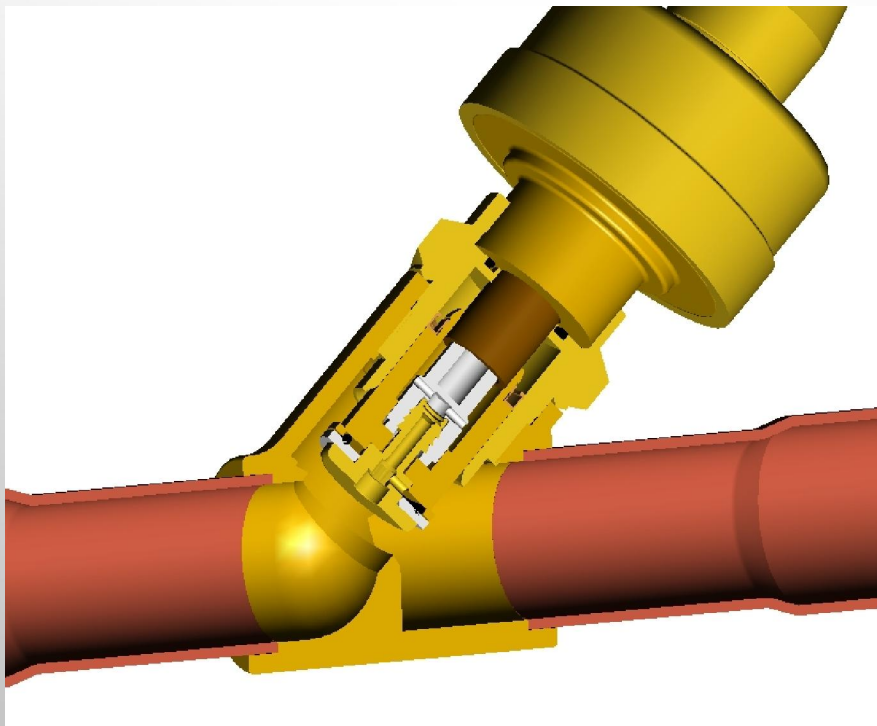
Steps / Sec.
Selector Sw.

Open / Close
Switch

Battery Power
Indicator



Stepper Valve



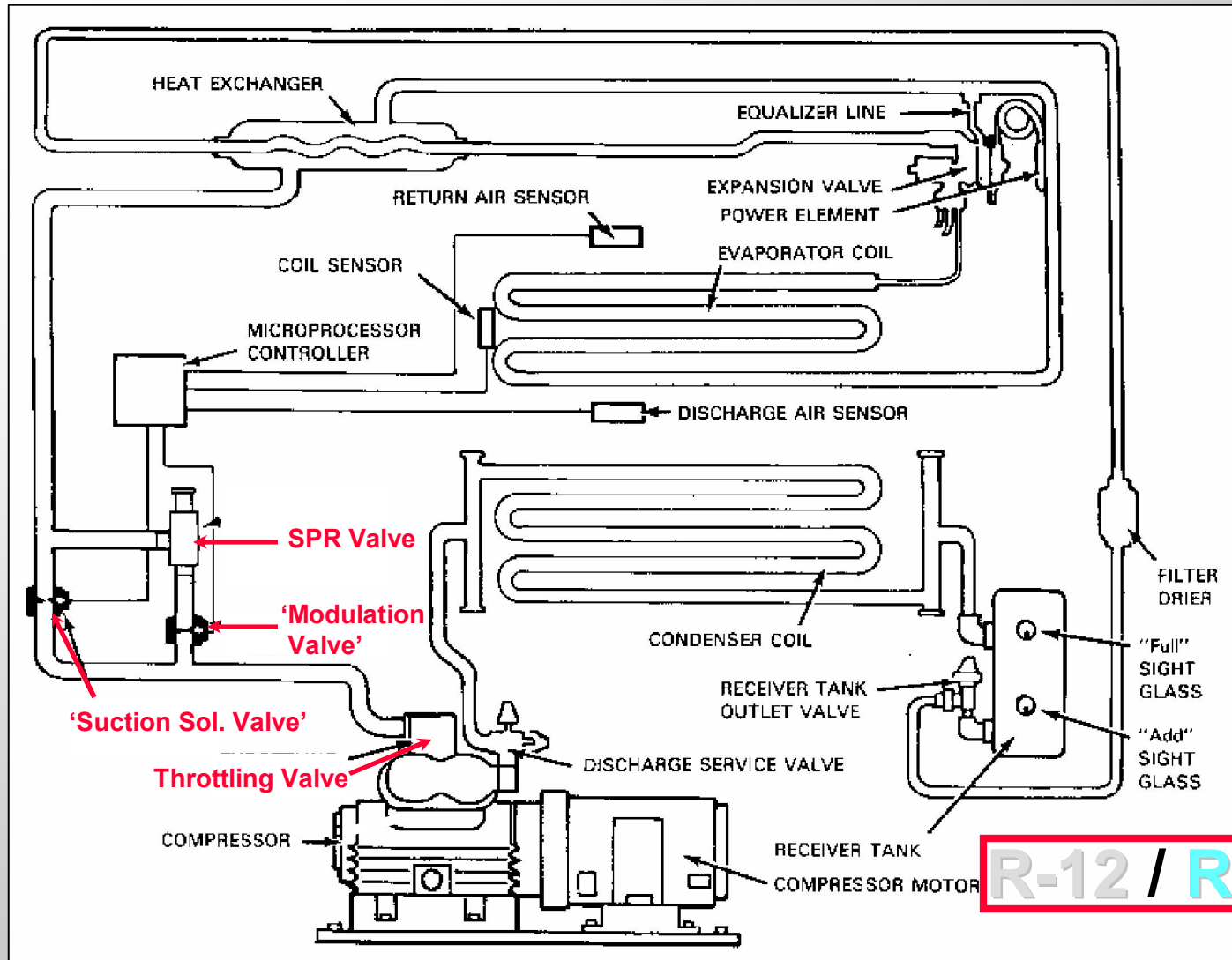


REFRIG. SYSTEMS

- CF-II / CF-III
- CRR
- CSR

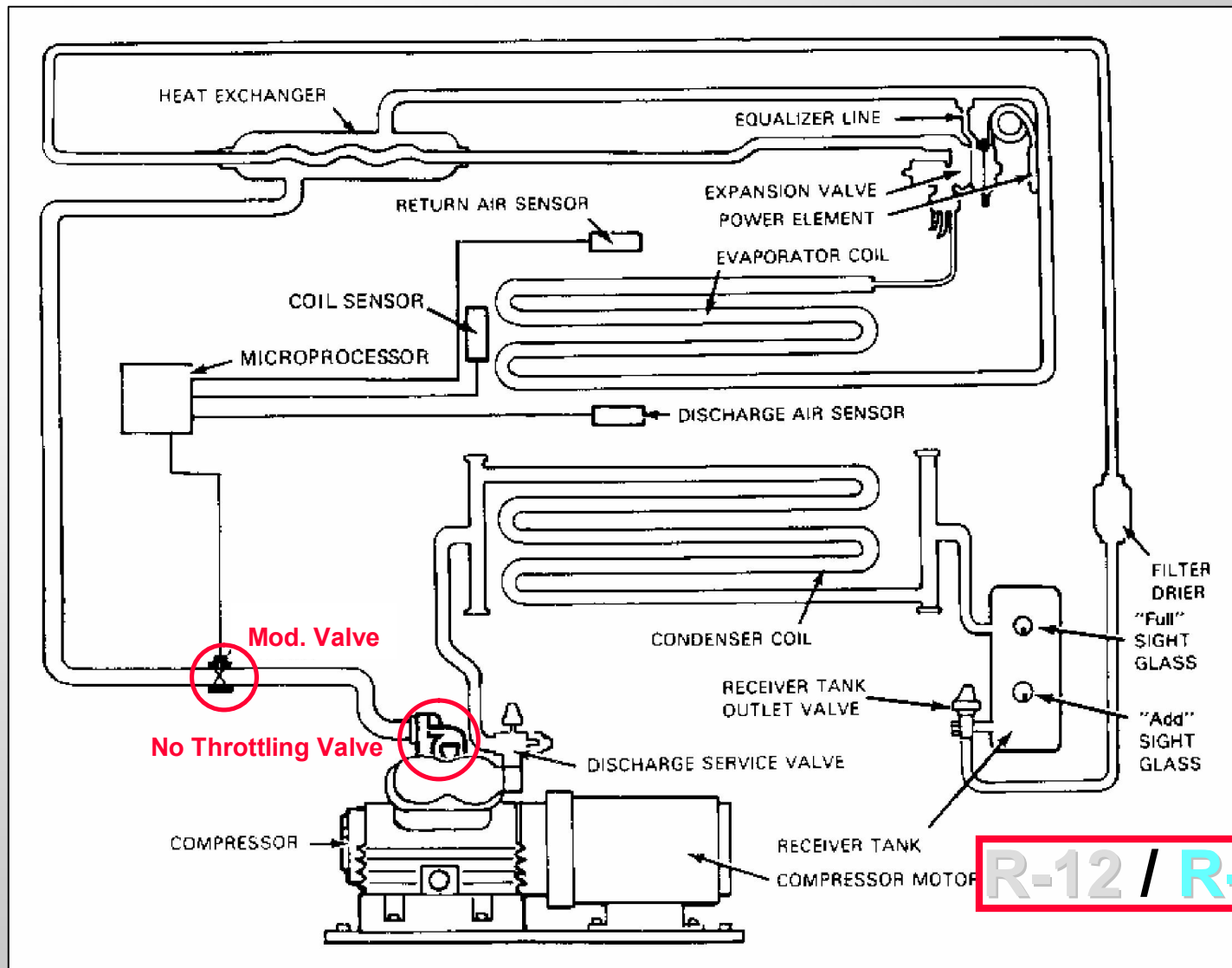


CF 'TWO VALVE' SYSTEM



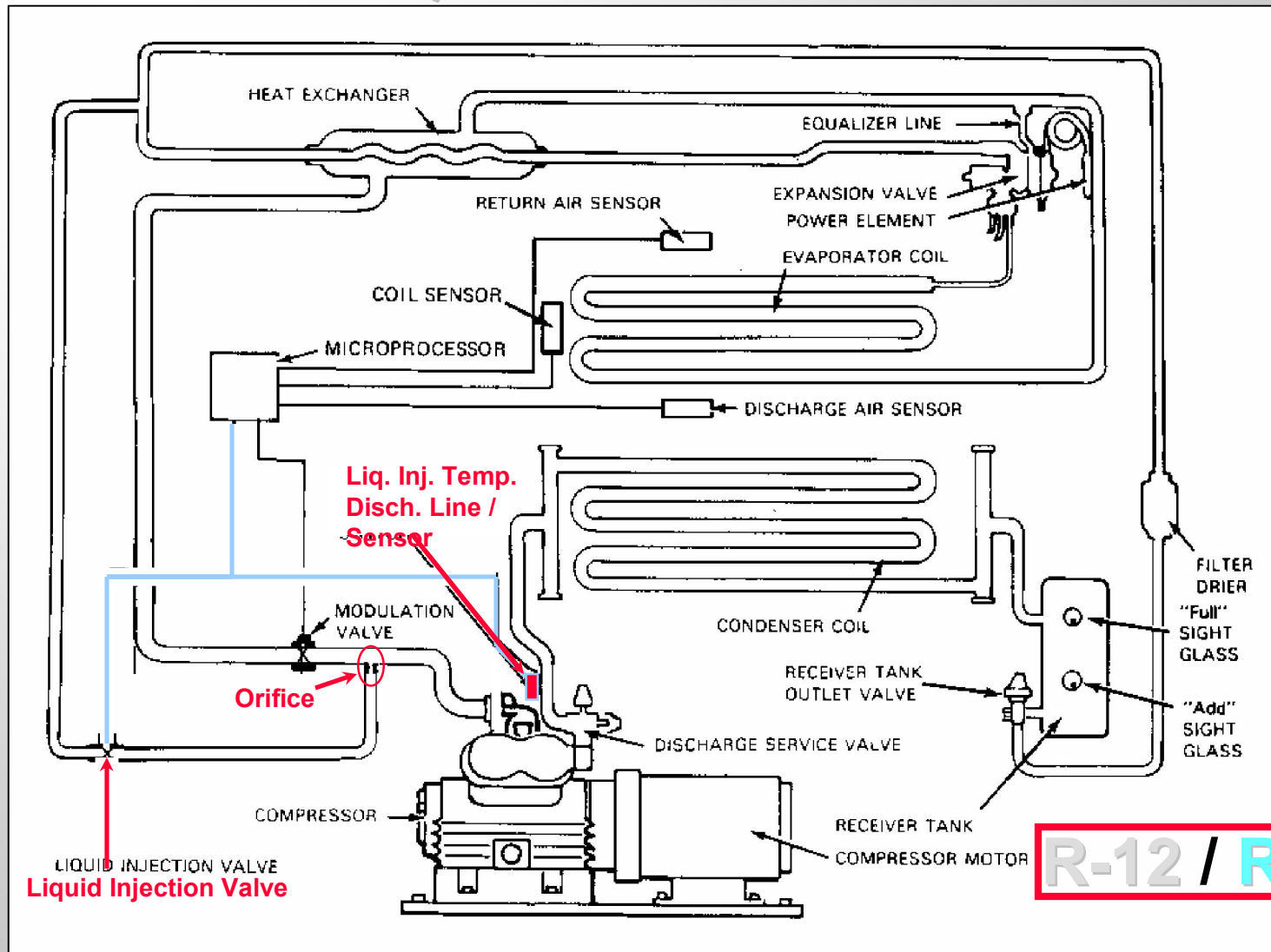


CF 'SINGLE VALVE' SYSTEM



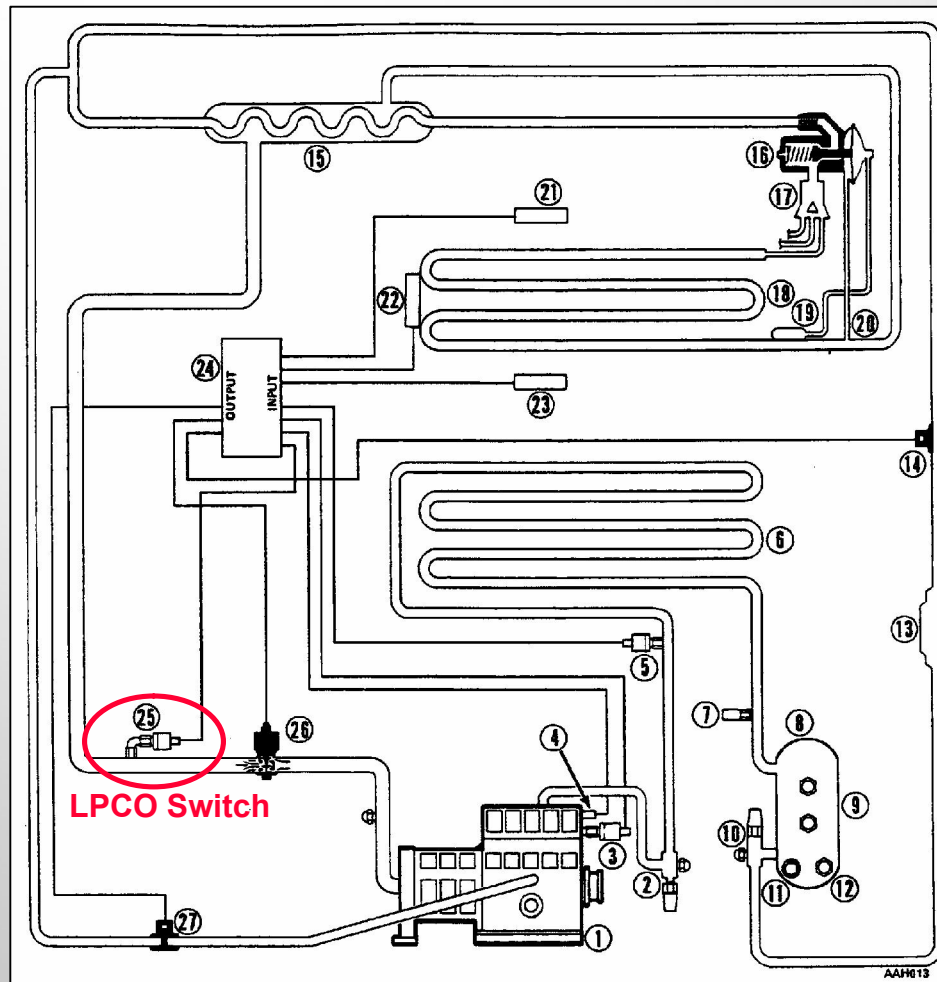


CF LIQUID INJ. SYSTEM





CF 3DS SYSTEM



Refrigeration Components

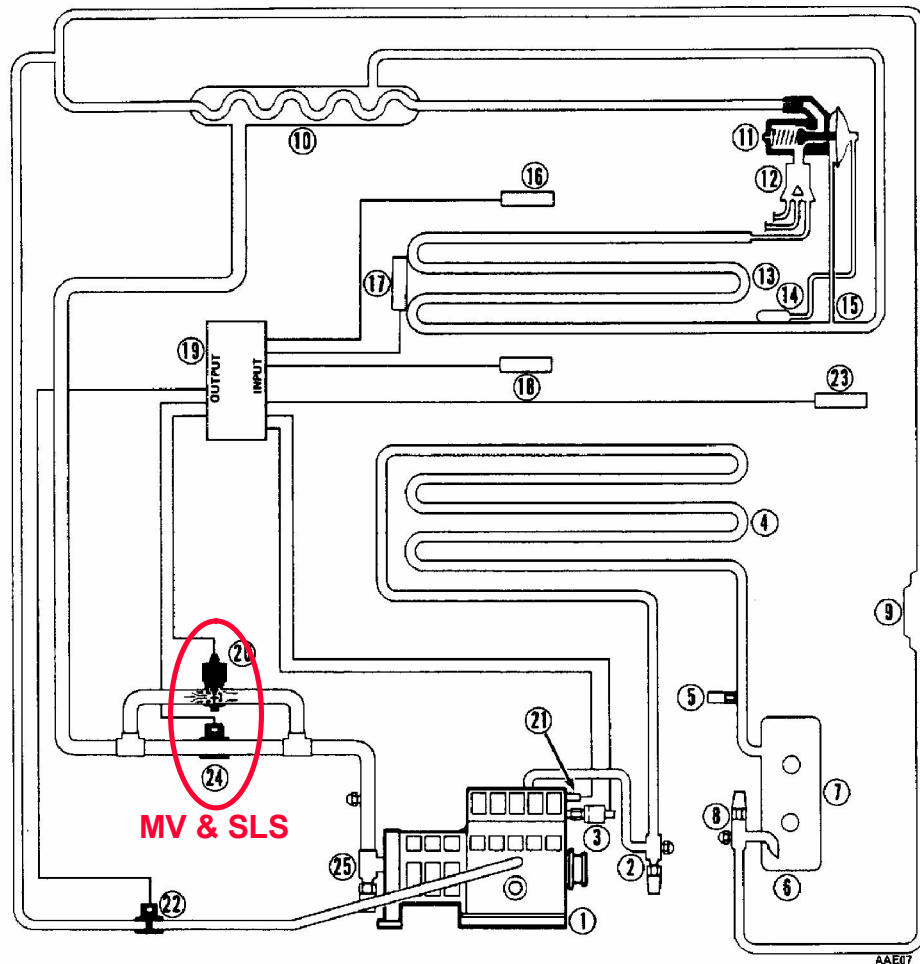
1. Compressor
2. Discharge Service Valve (DSV)
3. High Pressure Cutout Switch (HPCO)
4. Compressor Discharge Temperature Sensor
5. Condenser Fan Pressure Switch
6. Condenser Coil
7. High Pressure Relief Valve
8. Water-Cooled Condenser-Receiver Tank
9. Sight Glasses
10. Receiver Outlet Valve
11. Condenser Water Outlet
12. Condenser Water Inlet
13. Filter Drier
14. Liquid Line Solenoid (LLS)
15. Heat Exchanger
16. Expansion Valve (TXV)
17. Distributor
18. Evaporator
19. Power Element
20. Equalizer Line
21. Return Air Sensor
22. Coil Sensor
23. Discharge Air Sensor
24. Controller
25. Low Pressure Cutout Switch (LPCO)
26. Modulation Valve (MV)
27. Liquid Injection Valve (LIV)

R-12 / R-134a



CF-II M39 2DB SYSTEM

Refrigeration Schematic



Refrigeration Components

1. Compressor
2. Discharge Service Valve (DSV)
3. High Pressure Cutout Switch (HPCO)
4. Condenser
5. High Pressure Relief Valve
6. Receiver Tank
7. Sight Glasses
8. Receiver Outlet Valve
9. Filter Drier
10. Heat Exchanger
11. Expansion Valve (TXV)
12. Distributor
13. Evaporator
14. Power Element
15. Equalizer Line
16. Return Air Sensor
17. Coil Sensor
18. Discharge Air Sensor
19. Microprocessor Controller
20. Modulation Valve (MV)
21. Compressor Head (Discharge) Temperature Sensor
22. Liquid Injection Valve (LIV)
23. Ambient Sensor
24. Suction Line Solenoid (SLS)
25. Suction Service Valve (SSV)

R-404A



CRR-40 SYSTEM

Electrical, Refrigeration and MPC2000ID Menu Flow Diagrams

Refrigeration System Components

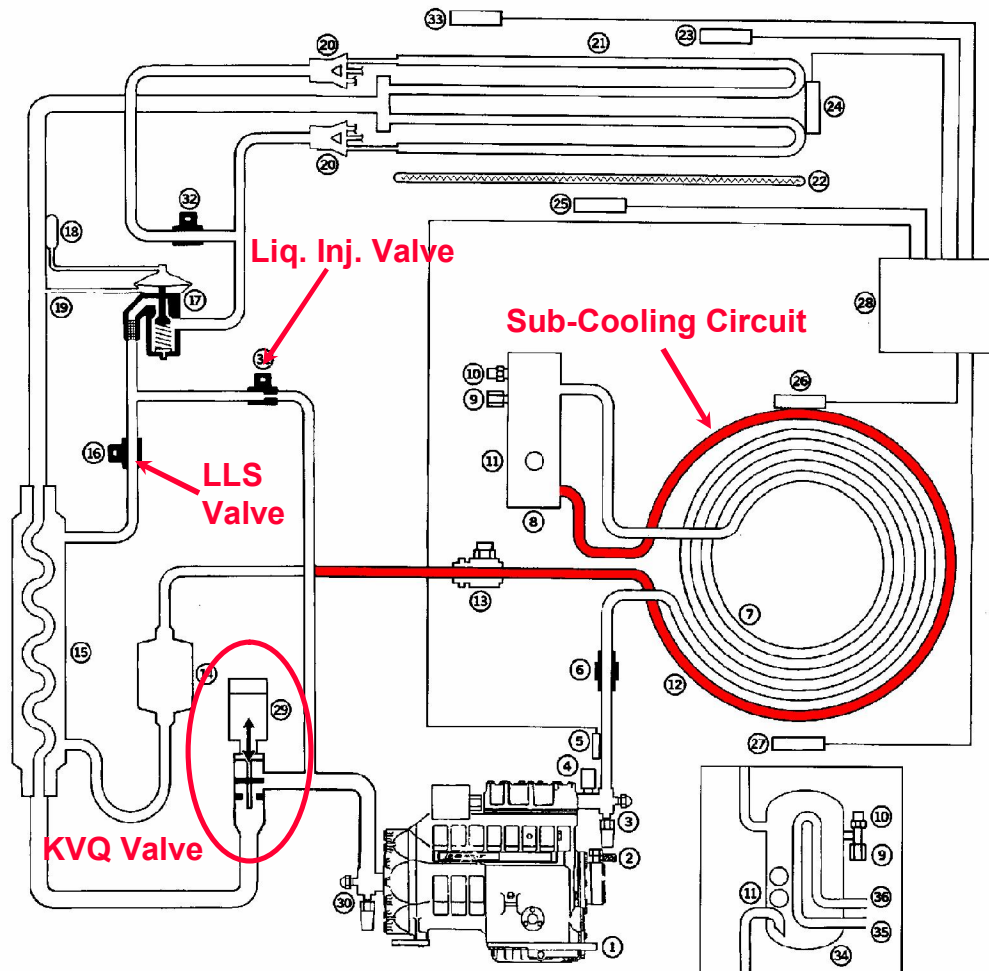
9-5

CRR-40

Refrigeration System Components

1. 3DS Compressor
2. Oil Fill / Drain Fitting
3. Discharge Service Valve
4. High Pressure Cutout Switch
5. Compressor Discharge Line Temperature Sensor
6. Condenser Check Valve
7. Condenser Coil (Circular)
8. Receiver Tank
9. High Pressure Relief (Fusible Plug)
10. Receiver Tank Service Fitting
11. Sight Glass(es)
12. Condenser Coil Subcooler Circuit
13. Liquid Line Service Valve
14. Dehydrator (Filter Drier)
15. Heat Exchanger
16. Liquid Line Solenoid (LLS)
17. Expansion Valve (TXV)
18. Expansion Valve Feeler Bulb
19. Equalizer Line
20. Distributor
21. Evaporator Coil
22. Electric Heaters
23. Return Air Sensor
24. Defrost (Evaporator Coil) Sensor
25. Supply Air Sensor
26. Condenser Coil Sensor
27. Ambient Sensor
28. Controller (MPC2000ID)
29. KVQ Valve (Evaporator Pressure Regulator)
30. Suction Service Valve
31. Liquid Injection Valve
32. Dehumidify Solenoid Valve (Option)
33. Humidity Sensor (Option)
34. Water-Cooled Condenser-Receiver Tank (Option)
35. Water Inlet Coupling
36. Water Outlet Coupling

R-134a



CRR-40, February 1999



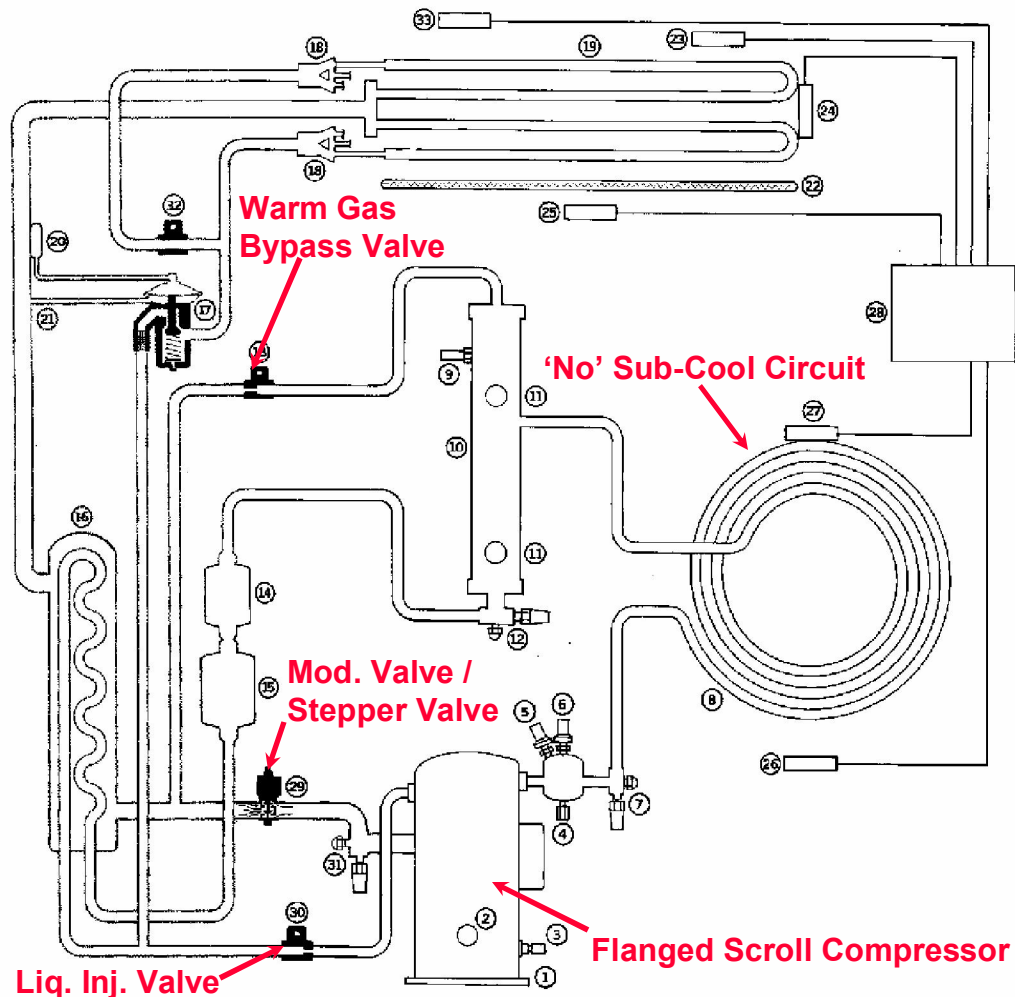
CSR-40 SYSTEM

CSR-40

Refrigeration System Components

1. Scroll Compressor
2. Sight Glass
3. Oil Fill / Drain Fitting
4. Compressor Head Discharge Temperature Sensor (CDTS)
5. High Pressure Cutout Switch (HPCO)
6. Condenser Fan Pressure Switch (CFPS)
7. Discharge Service Valve (DSV)
8. Condenser Coil (Circular)
9. High Pressure Relief Valve
10. Receiver Tank
11. Receiver Tank Sight Glasses
12. Receiver Tank Outlet Valve
13. Receiver Bypass Valve
14. Compressor Oil Filter
15. Filter Drier (Dehydrator)
16. Heat Exchanger
17. Expansion Valve (TXV)
18. Distributors
19. Evaporator Coil
20. Expansion Valve Feeler Bulb
21. Equalizer Line
22. Electric Heaters
23. Return Air Sensor (RET)
24. Evaporator Coil Sensor (ECOIL)
25. Supply Air Sensor (SUP)
26. Ambient Sensor (AMBT)
27. Condenser Coil Sensor (CCOIL)
28. Controller (µP-D)
29. Modulation Valve (MV)
30. Liquid Injection Valve (LIV) with Restrictor
31. Suction Service Valve (SSV)
32. Dehumidify Solenoid Valve (DSV) (Option)
33. Humidity Sensor (rH) (Option)

R-404A



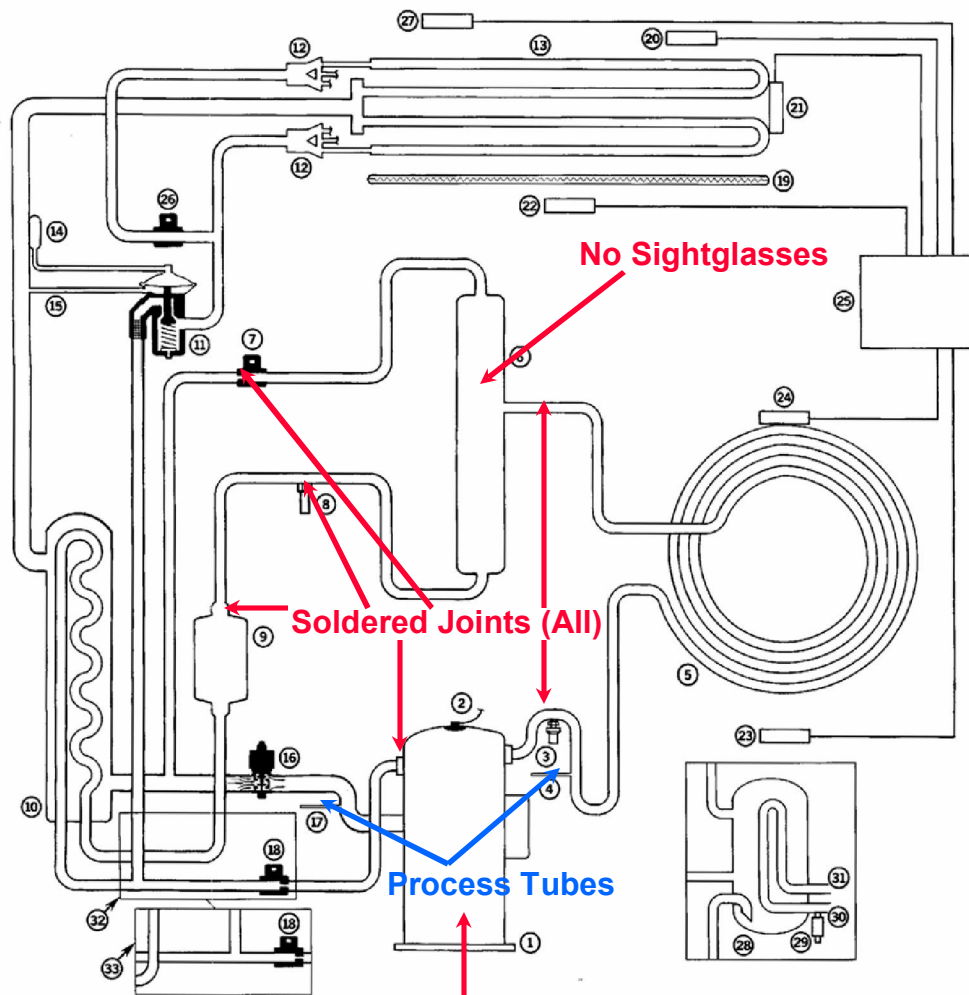


CSR-20/40 HERMETIC SYSTEM

CSR-20 & CSR-40 Refrigeration System Components

1. Scroll Compressor
2. Compressor Discharge Temperature Sensor (CDTS)
3. High Pressure Cutout Switch (HPCO)
4. Discharge Line Process Tube
5. Condenser Coil (Circular)
6. Receiver Tank
7. Warm Gas Bypass Valve
8. High Pressure Relief Valve
9. Filter Drier/In-line Filter
10. Heat Exchanger
11. Expansion Valve (TXV)
12. Distributors
13. Evaporator Coil
14. Expansion Valve Feeler Bulb
15. Equalizer Line
16. Modulation Valve (MV)
17. Suction Line Process Tube
18. Liquid Injection Valve (LIV) with Restrictor
19. Electric Heaters
20. Return Air Sensor (RET)
21. Evaporator Coil Sensor (ECOIL)
22. Supply Air Sensor (SUP)
23. Ambient Sensor (AMBT)
24. Condenser Coil Sensor (CCOIL)
25. Controller (µP-D)
26. Dehumidify Solenoid Valve (DSV) (Option)
27. Humidity Sensor (RH) (Option)
28. Water-Cooled Condenser-Receiver Tank (Option)
29. Water Pressure Switch (WPS) (Option)
30. Water Inlet Coupling (Option)
31. Water Outlet Coupling (Option)
32. Units Built After 7/99: Liquid Line and Liquid Injection Valve Piping
33. Units Built Before 7/99: Liquid Line and Liquid Injection Valve Piping

R-404A



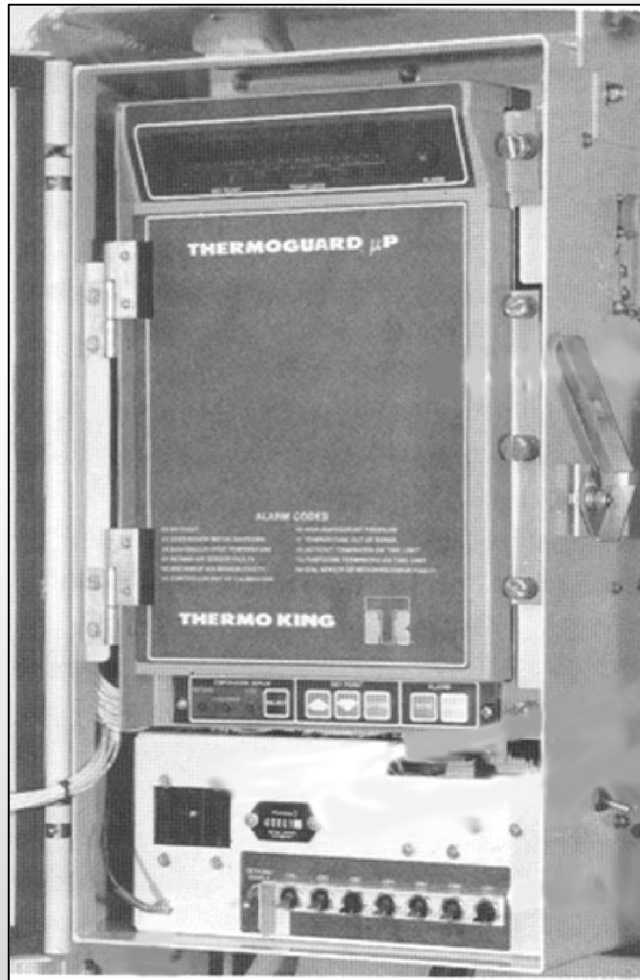
Hermetic Scroll



CONTROLLERS

Models

- μ P





CONTROLLERS

Models

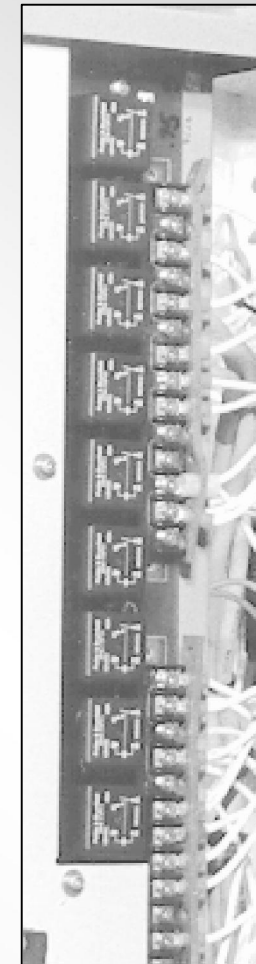
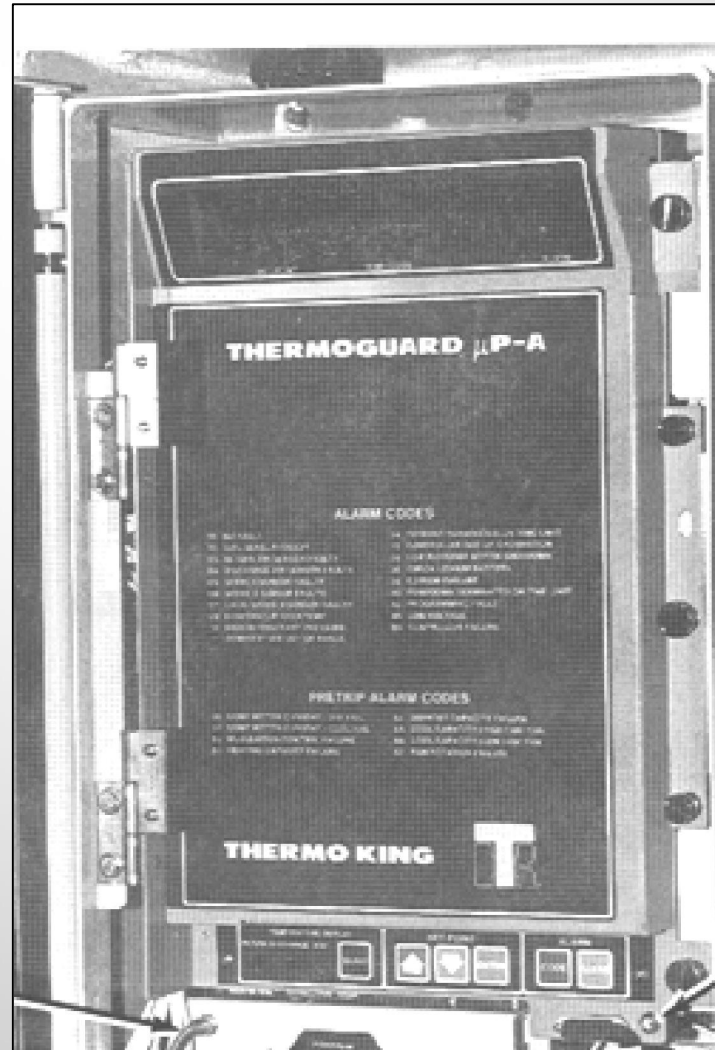
- μP
- μC



CONTROLLERS

Models

- μP
- μC
- $\mu P-A$





CONTROLLERS

Models

- μP
- μC
- $\mu P-A$
- $\mu P-A+$

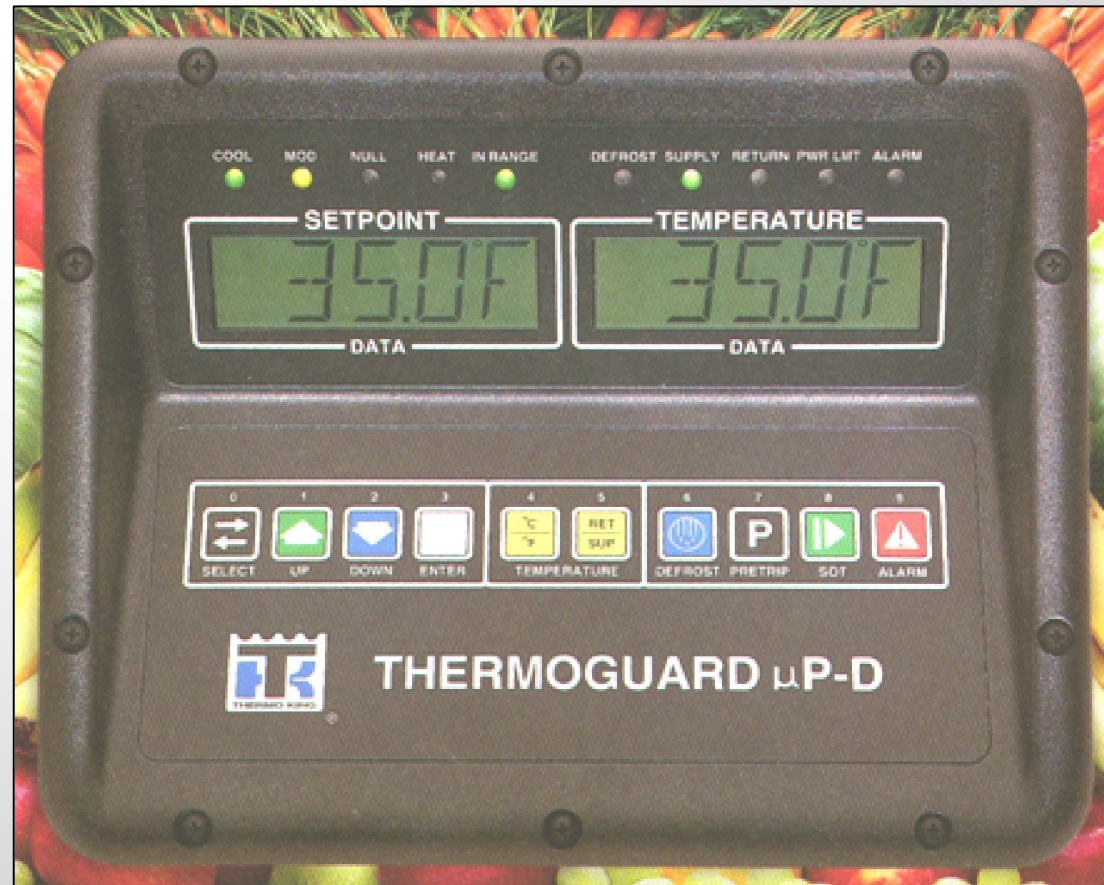




CONTROLLERS

Models

- μP
- μC
- $\mu P-A$
- $\mu P-A+$
- $\mu P-D$

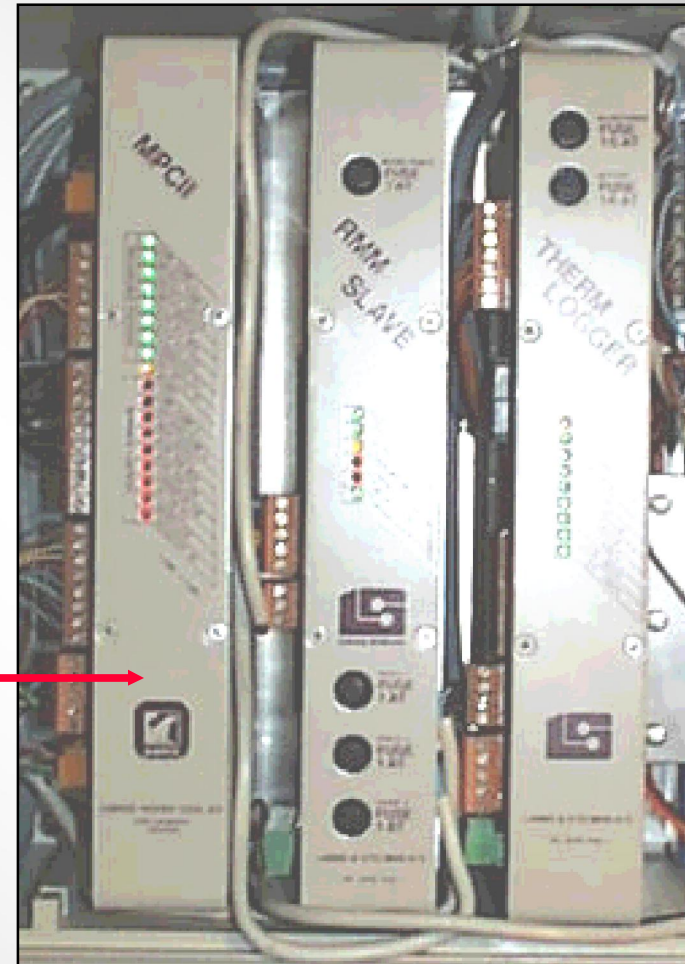
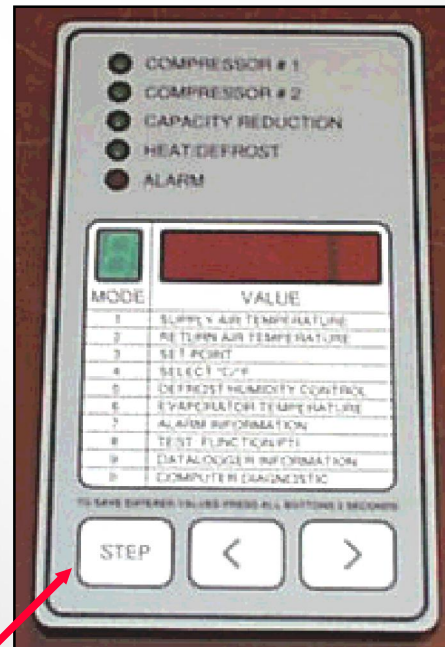




CONTROLLERS

Models

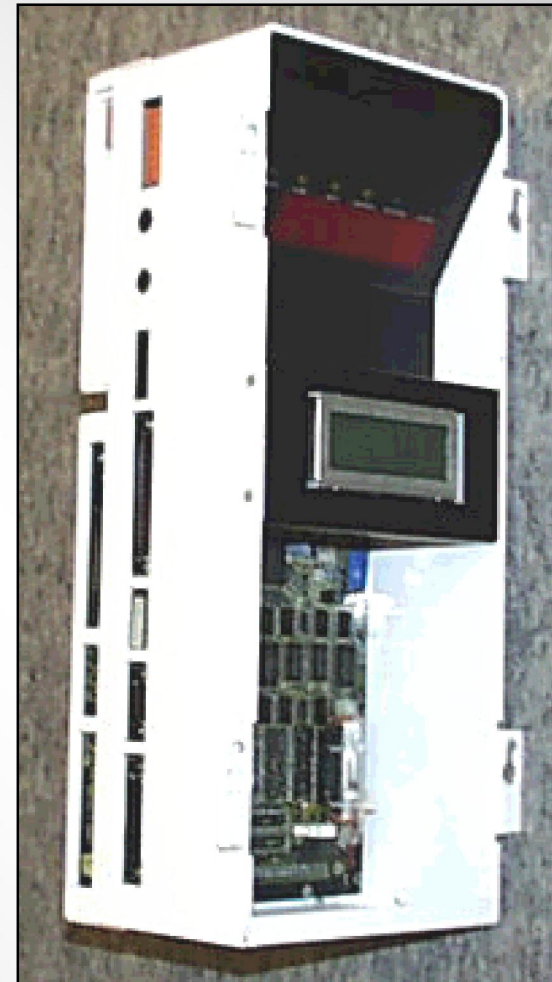
- μ P
- μ C
- μ P-A
- μ P-A+
- μ P-D
- **MPC2**



CONTROLLERS

Models

- μP
- μC
- $\mu P-A$
- $\mu P-A+$
- $\mu P-D$
- MPC2
- **MPC2000**

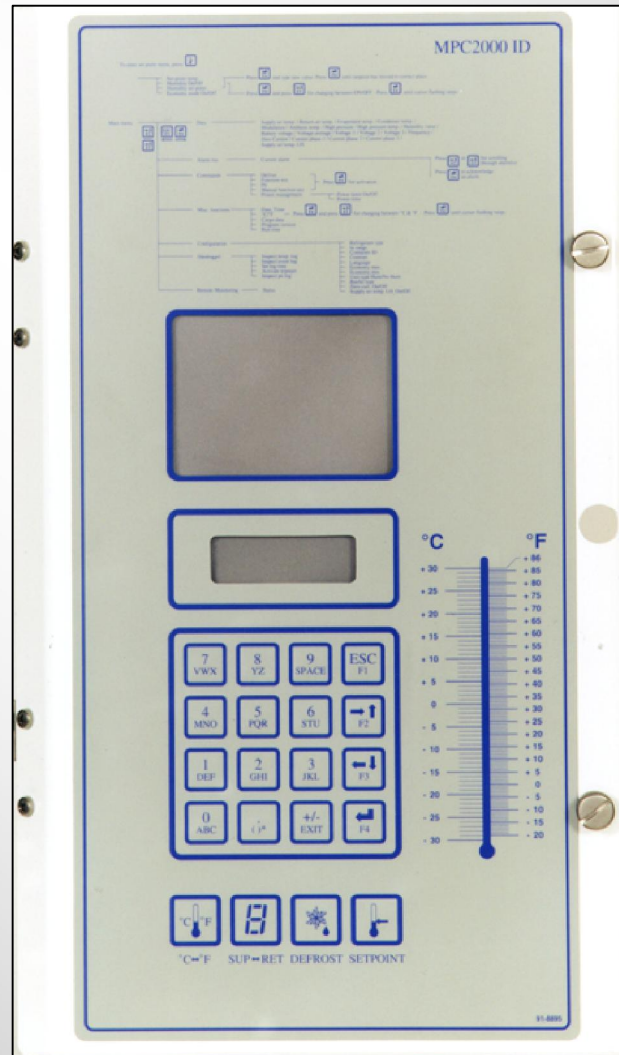




CONTROLLERS

Models

- μ P
- μ C
- μ P-A
- μ P-A+
- μ P-D
- MPC2
- MPC2000
- **MPC2000ID**

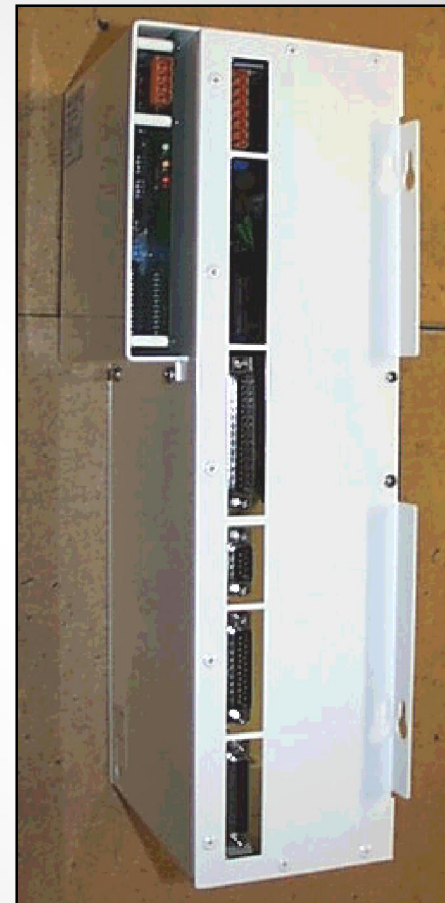
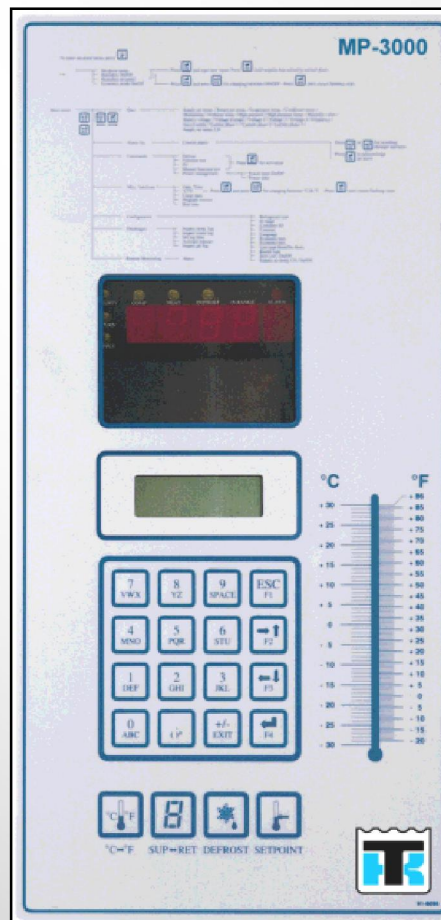




CONTROLLERS

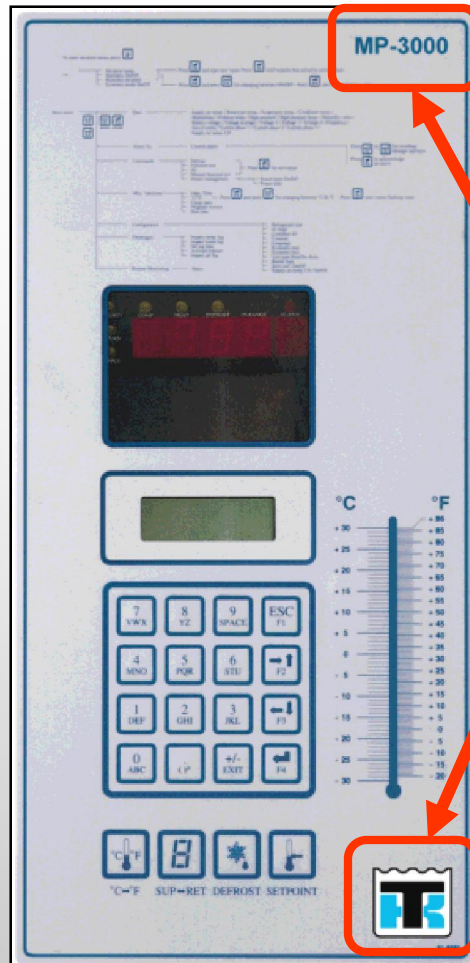
Models

- μ P
- μ C
- μ P-A
- μ P-A+
- μ P-D
- MPC2
- MPC2000
- MPC2000ID
- **MP-3000**





MP-3000 CONTROLLER



Now 'Standard'

New Decal

Stepper Motor
Control



MP-3000 CON'T



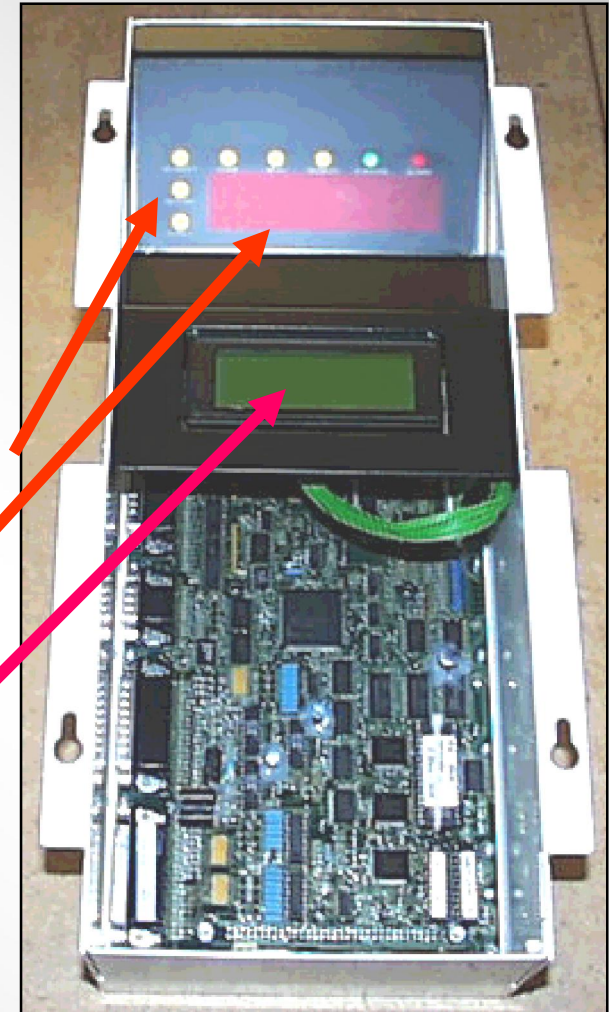
Optional RMM

Mode Indicators

LED Display

LCD Display

Unit Config.
Switch

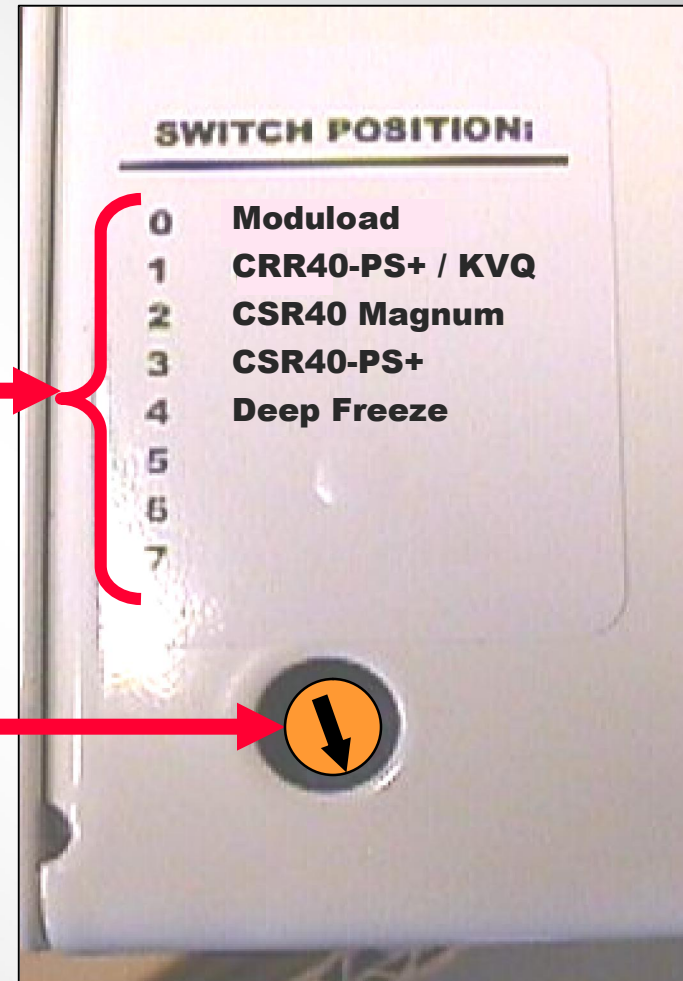




UNIT CONFIG. SWITCH

**Switch Position /
Unit Description**

Switch Location





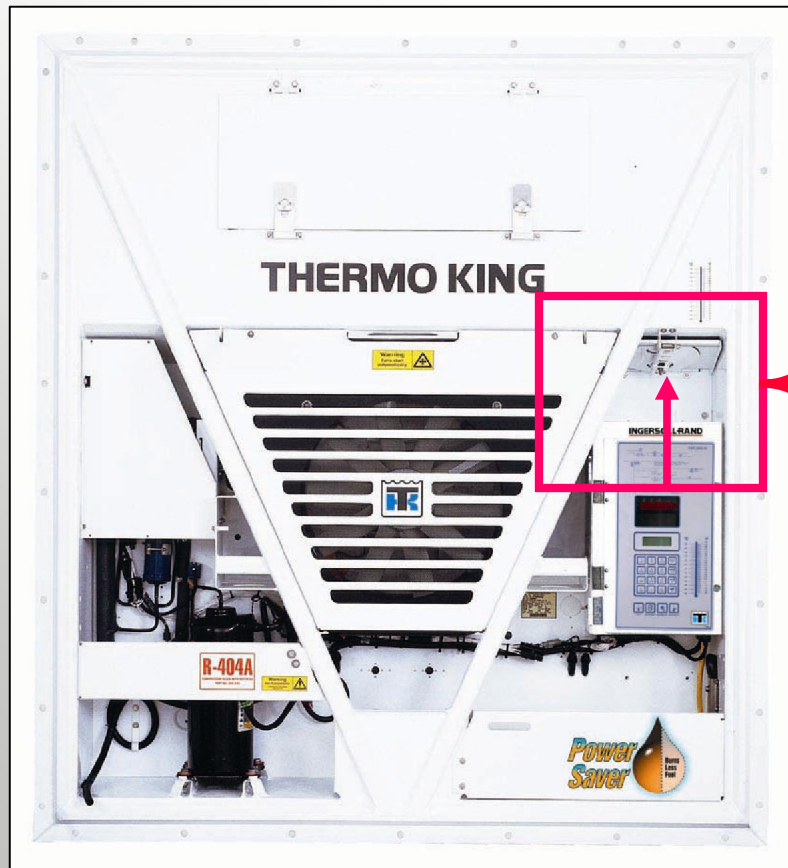
OPTIONS

- **AFAM / AFAM +**
- **WATER COOLED CONDENSER**



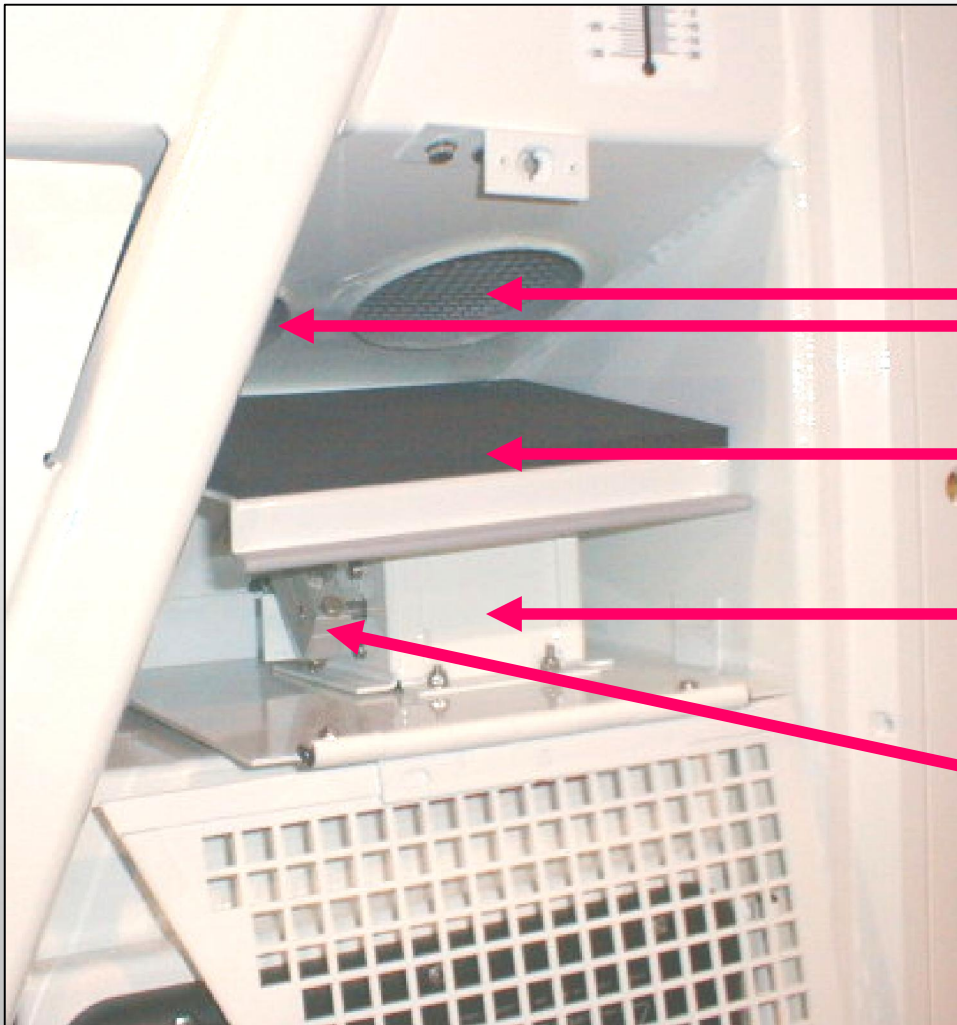
AFAM - OPTION

Advanced Fresh Air Management





AFAM CON'T



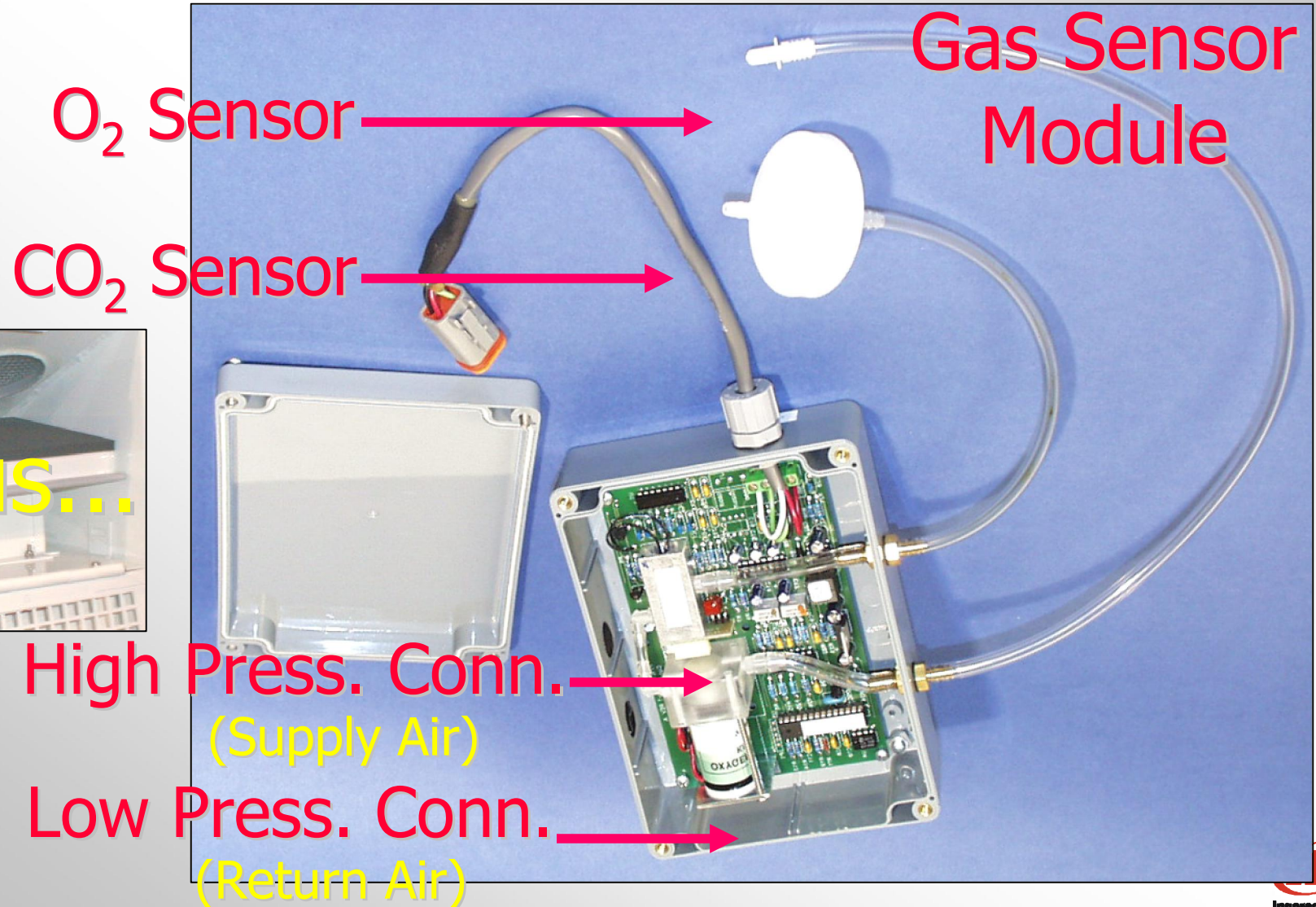
Air Exchange Ports

Hinged Door

Servo-Motor / Encl.

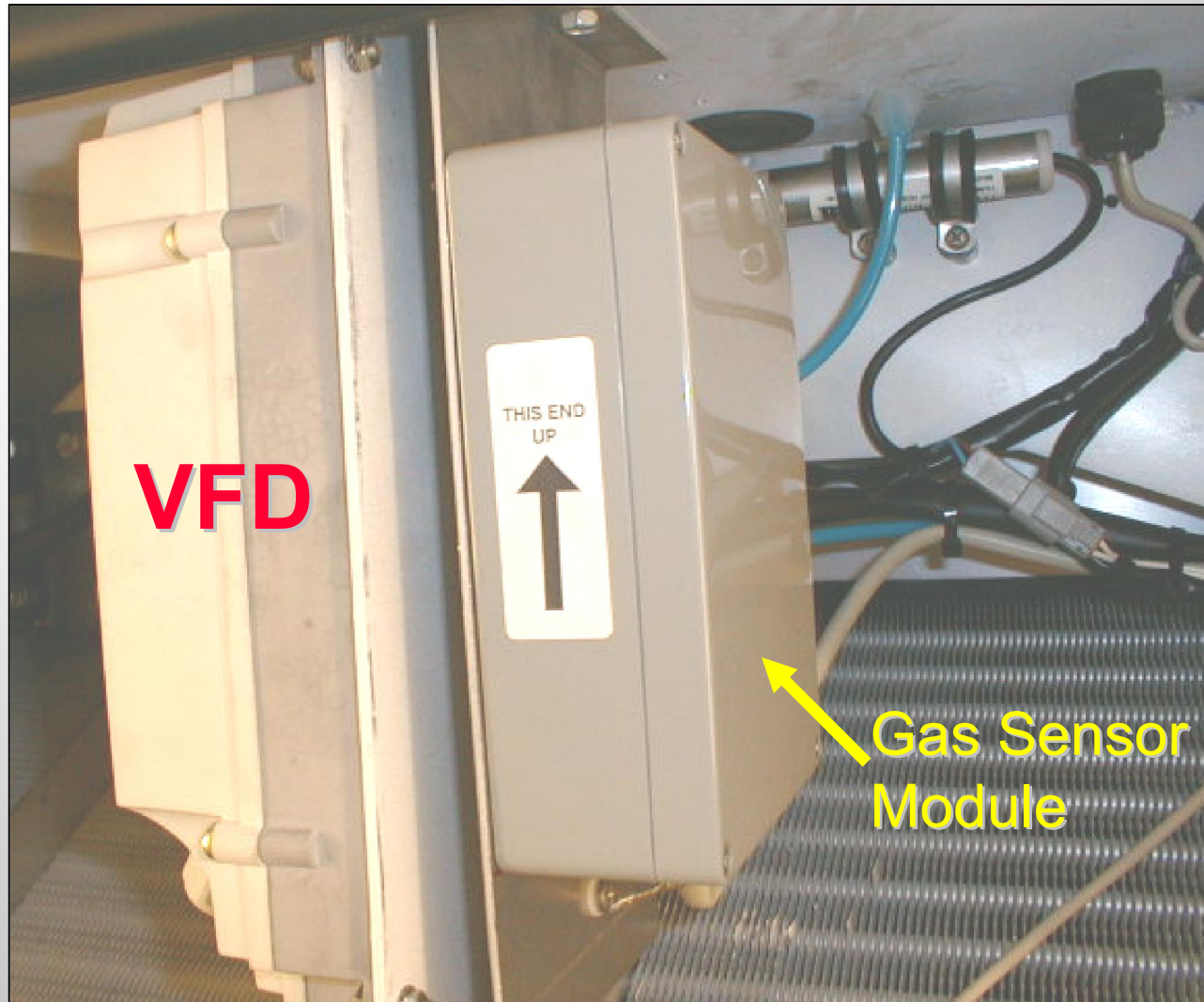
Shaft / Linkage

AFAM '+'





GAS SENSOR LOCATION



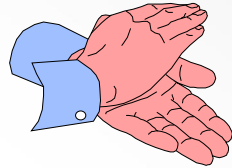
VFD

**Gas Sensor
Module**



WATER-COOLED CONDENSER





THE 'BOW'





Container Training

